MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT MISCELLANEOUS STRUCTURE

DESIGN CALCULATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2023

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
Employer	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works		Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The document is prepared for Design Calculation of Foundation for Miscellaneous Equipments in Power Plant.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The document is prepared for Design Calculation of Miscellaneous Equipment Foundation for Power Plant. This includes Boiler, Boiler Washing Tank, Condensate Return Tank, Heat Recovery Container, PACU Unit & Black Start DG. All the loads applied for analysis & design followed Equipment datasheets as attached.

Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008

Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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4 DESIGN OF MISCELLANEOUS FOUNDATIONS

4.1 Design of Boiler Foundation (Refer Exhaust Gas boiler Drawing No. 8159-69M)

	145			146			220			222		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	0.0	-3.3	0.0	0.0	-3.3	0.0	0.0	-3.3	0.0	0.0	-3.3	0.0
2 LL	0.9	-1.3	0.0	0.0	-1.3	0.0	0.0	-1.3	0.0	0.0	-1.3	0.0
3 WL(+X)	0.5	1.5	0.0	0.5	1.5	0.0	0.5	-1.5	0.0	0.5	-1.5	0.0
4 WL(+Z)	0.0	1.5	0.5	0.0	-1.5	0.5	0.0	1.5	0.5	0.0	-1.5	0.5
5 SL(+X)	1.0	2.5	0.0	1.0	2.5	0.0	1.0	-2.5	0.0	1.0	-2.5	0.0
6 SL(+Z)	0.0	2.5	1.0	0.0	-2.5	1.0	0.0	2.5	1.0	0.0	-2.5	1.0

(Refer Heat Recovery Container Drawing No. 8159-69M)

TOTAL DEAD LOAD ON 4 LEGS OF BOILER =		13.2	T	
DEAD LOAD ON EACH LEG =	13.2/4 =	3.3	T	
TOTAL LIVE LOAD ON 4 LEGS OF CONTAINER =		5.2	T	(18.4-13.2) = 5.2T
LIVE LOAD ON EACH LEG =	5.2/4 =	1.3	T	

WIND LOAD:

INTENSITY OF WIND LOAD FOR POWER PLANT AREA =		0.13	T/M^2	(REFER CALC. NO. MD-216-49PR-EG-CV-CAL-2014)
TOTAL WIND LOAD =	0.13*2.364*6.4 =	2.0	T	
MOMENT AT BASE =	1.966848 * 6.4/2 =	7.0	T-M	
SHEAR PER LEG =		0.5	T	
MAX. COMP./TENSILE FORCE PER LEG =	7/ (2*2.364) =	1.5	T	

EARTHQUAKE LOAD :

SEISMIC LOAD = 0.02g OF TOTAL WEIGHT (say) =		3.6	T	
MOMENT AT BASE =	3.61008 * 6.4/2 =	12.0	T-M	
SHEAR PER LEG =		0.9	T	
MAX. COMP./TENSILE FORCE PER LEG =	12/(2*2.364)=	2.5	T	

FOUNDATION SIZE (F1) =	2.75m	X	2.75m	
DEPTH OF THE FOUNDATION =	1.50m		from FGL	
THE WT OF SOIL +SELF WT	2.75x2.75x1.5x2 =		22.6875	T

WORKING STRESS DESIGN	145			146			220			222		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	0.88	-4.55	0.00	0.00	-4.55	0.01	0.00	-4.55	0.00	0.00	-4.55	0.02
DL+0.75*LL+0.75*WLX	1.03	-3.11	0.00	0.38	-3.11	0.01	0.38	-5.36	0.00	0.37	-5.36	0.02
DL+0.75*LL+0.75*WLZ	0.66	-3.11	0.38	0.00	-5.36	0.38	-0.01	-3.11	0.37	-0.02	-5.36	0.39
DL+0.75*LL+0.75*0.7*SLX	1.18	-2.93	0.00	0.53	-2.93	0.02	0.53	-5.55	0.00	0.52	-5.55	0.02
DL+0.75*LL+0.75*0.7*SLZ	0.66	-2.93	0.53	0.00	-5.55	0.53	0.00	-2.93	0.52	0.00	-5.55	0.54
0.6*DL+WLX	0.50	-0.48	0.00	0.50	-0.48	0.00	0.50	-3.48	0.00	0.50	-3.48	0.00
0.6*DL+WLZ	0.00	-0.48	0.50	0.00	-3.48	0.50	-0.01	-0.48	0.50	-0.02	-3.48	0.50
0.6*DL+0.7*SLX	0.70	-0.23	0.00	0.70	-0.23	0.01	0.70	-3.73	0.00	0.70	-3.73	0.01
0.6*DL+0.7*SLZ	-0.01	-0.23	0.70	0.01	-3.73	0.70	0.00	-0.23	0.70	0.00	-3.73	0.70

SLIDING AND OVERTURNING CHECK

Load at bottom of foundation					OVERTURNING CHECK			SLIDING CHECK, μ = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	40.89	0.9	0.1	0.03	1.49	37.70	1102.36	18.64	545.17
DL+0.75*LL+0.75*WLX	39.64	2.2	0.0	0.02	7.49	7.27	1444.13	7.35	714.19
DL+0.75*LL+0.75*WLZ	39.64	0.6	6.4	1.52	1.08	50.46	8.50	24.95	10.42
DL+0.75*LL+0.75*0.7*SLX	39.64	2.8	0.1	0.03	9.15	5.96	921.26	5.75	455.60
DL+0.75*LL+0.75*0.7*SLZ	39.64	0.7	8.1	2.12	1.12	48.73	6.75	24.10	7.47
0.6*DL+WLX	30.61	2.0	0.0	0.00	8.50	4.95	34383.43	6.12	17004.17
0.6*DL+WLZ	30.61	0.0	8.5	2.00	0.05	825.20	4.95	408.10	6.12
0.6*DL+0.7*SLX	30.61	2.8	0.0	0.02	10.71	3.93	1539.56	4.37	761.38
0.6*DL+0.7*SLZ	30.61	0.0	10.7	2.80	0.00	-	3.93	N.A	4.37

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS $50T/m^2$

MAXIMUM GROSS BEARING PRESSURE OBTAINED $9T/m^2$ IN LC 105 (DL+0.75LL+0.75*0.7SL(+X). HENCE IT IS SAFE.

REFER TO THE STAAD FILE ATTACHED.

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Foundation has been designed with the Design strength Load Combinations of Staad File (LC 200 - LC 217)

REINFORCEMENT CALCULATION OF FOUNDATION:

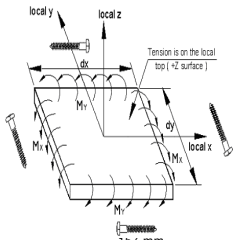
Foundation Thickness =	250	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.916	T-M	Corresponding Mxy	-0.20	T-M	*(Plate No. - 349 , L/C No. - 212)*
Min. Mx	-1.498	T-M	Corresponding Mxy	0.01	T-M	*(Plate No. - 611 , L/C No. - 213)*
Max. My	0.916	T-M	Corresponding Mxy	-0.20	T-M	*(Plate No. - 375 , L/C No. - 210)*
Min. My	-1.498	T-M	Corresponding Mxy	0.01	T-M	*(Plate No. - 94 , L/C No. - 211)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	0.916 + 0.199 =	1.115 T-M
My1 =	My + abs(Mxy) =	0.916 + 0.199 =	1.115 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	0.916 + 0.043 =	0.959 T-M
My2 =	My+abs(Mxy^2/Mx) =	0.916 + 0.043 =	0.959 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 1.115 T-m
Myd = 1.115 T-m

ALONG X - DIRECTION :

Mu = 1.115 T-m
Φ*Kn = Mu/bd^2 deff. = 157 mm
Mu/bd^2 = 1.115X10^7/(1000X157^2) = 0.452 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0.0012X1000X157/100 = 1.88 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.115 T-m
Φ*Kn = Mu/bd^2 deff. = 169 mm
Mu/bd^2 = 1.115X10^7/(1000X169^2) = 0.39 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X169/100 = 1.69 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-1.498 - 0.009 =	-1.507 T-M
My1 =	My - abs(Mxy) =	-1.498 - 0.009 =	-1.507 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-1.498 - 0 =	-1.498 T-M
My2 =	My-abs(Mxy^2/Mx) =	-1.498 - 0 =	-1.498 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -1.507 T-M
Myd = -1.507 T-M

ALONG X - DIRECTION :

Mu = 1.507 T-m
Φ*Kn = Mu/bd^2 deff. = 157 mm
Mu/bd^2 = 1.507X10^7/(1000X157^2) = 0.611 N/mm^2 < 6.9 N/mm^2
Pt = 0.0016 Reqd. Ast = 0.0016X1000X157/100 = 2.51 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 1.507 T-m
Φ*Kn = Mu/bd^2 deff. = 169 mm
Mu/bd^2 = 1.507X10^7/(1000X169^2) = 0.528 N/mm^2 < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X169/100 = 2.37 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

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SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.129 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 651 , L/C No. - 210)*

Vu = 2.025 T

Provided Pt = 0.0036

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0036X2.025X10^4X157/1.507/10^7)X1000X157/10^4

= 15.064 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0036)X1000X157/10^4

= 15.822 T

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X157/10^4

= 26.936 T

HENCE, Vc = 15.064 T

Φt*Vc = 0.75X15.064 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 11.298 T **> 2.025T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.227 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 651 , L/C No. - 213)*

Vu = 3.836 T

Pt provided = 0.0033

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0033X3.836X10^4X169/1.507/10^7)X1000X169/10^4

= 16.405 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0033)X1000X169/10^4

= 16.945 KN

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X169/10^4

= 28.995 T

HENCE, Vc = 16.405 T

Φt*Vc = 0.75X16.405 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 12.304 T **> 3.836T , Hence O.K.**

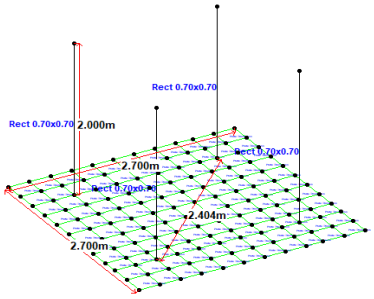
DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F1

Size of Column = 700 X 700 mm^2

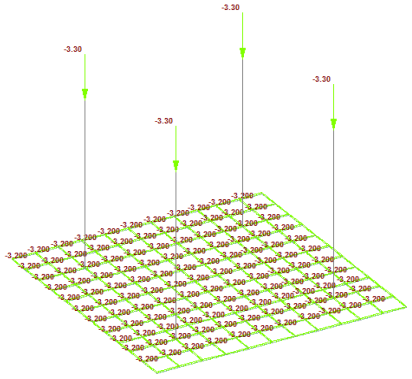
Reinforcement Provided = 16 no. 16 dia bars

Area of Reinforcement Provided = 3217.0 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation =		1 T/m^2
Soil Weight over the foundation =	1.2x 1.8 T/m2 =	2.16 T/m^2
SOIL+SURCHARGE LOAD =		3.16 T/m^2

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Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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4.2 Design of Foundation for Heat Recovery Container

Load Detail	Each leg of Base Frame		
	FX (T)	FY (T)	FZ (T)
1 DL	0.0	3.9	0.0
2 LL	0.0	2.4	0.0
3 WL1(+X)	4.3	13.1	0.0
4 WL2(-X)	-4.3	-6.5	0.0
5 EL1(+X)	2.0	9.0	0.0
6 EL2(-X)	-2.0	-2.4	0.0

(Refer Heat Recovery Container Drawing No. 8159-94M)

TOTAL DEAD LOAD ON 4 LEGS OF CONTAINER =		15.5	T	
DEAD LOAD ON EACH LEG =	15.5/4 =	3.9	T	
TOTAL LIVE LOAD ON 4 LEGS OF CONTAINER =		9.5	T	(25-15.5) = 9.5T
LIVE LOAD ON EACH LEG =	9.5/4 =	2.4	T	
WIND LOAD:				
SHEAR PER LEG =		4.3	T	
MAX. COMP. FORCE PER LEG =	19.1/(2.45*2) + 9.2 =	13.1	T	
MAX. TENS. FORCE PER LEG =	19.1/(2.45*2) + 2.6 =	6.5	T	
EARTHQUAKE LOAD IN X DIR (ACROSS BASE FRAME):				
SHEAR PER LEG =		2.0	T	
MAX. COMP. FORCE PER LEG =	11.8/(2.45*2) + 6.6 =	9.0	T	
MAX. TENS. FORCE PER LEG =	11.8/(2.45*2) =	2.4	T	

HEAT RECOVERY CONTAINER is supported on a Base Frame at bottom. Each Foundation beam of width 500mm & depth 1000mm has been designed along the base frame of equipment supporting 2 legs of the Container. Width of Foundation beam is increased to 1m at base to assure stability of the foundation system.

FOUNDATION DATA

	ABOUT-X	ABOUT-Z
Size of fdn. raft :	0.50m	12.59m
Size of fdn. base:	1.00m	12.59m
Thickness of fdn beam :	1.00 m	
Thickness of fdn base :	0.20 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-0.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

Calculation of Weight of Foundation :

Weight of Foundation beam = (0.5*12.59*1*2.5)= 15.74T
Weight of Foundation base = 6.30T
22.03T

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

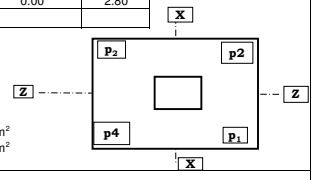
WORKING STRESS DESIGN	Load at U/S Of Base Plt			Load at bottom of foundation						
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
D+L	0.00	12.50	0.00	0.0	0.00	34.53	0.00	0.00	0.00	0.00
DL+0.75*L+0.75*WL1	6.45	30.96	0.00	0.0	0.00	53.00	6.45	0.00	0.00	6.45
DL+0.75*L+0.75*WL2	-6.45	1.56	0.00	0.0	0.00	23.60	-6.45	0.00	0.00	6.45
DL+0.75*L+0.75*0.7*EL1	2.10	20.76	0.00	0.0	0.00	42.80	2.10	0.00	0.00	2.10
DL+0.75*L+0.75*0.7*EL2	-2.10	8.79	0.00	0.0	0.00	30.83	-2.10	0.00	0.00	2.10
0.6*DL+WL1	8.60	30.85	0.00	0.0	0.00	52.88	8.60	0.00	0.00	8.60
0.6*DL+WL2	-8.60	-8.35	0.00	0.0	0.00	13.68	-8.60	0.00	0.00	8.60
0.6*DL+0.7*EL1	2.80	17.25	0.00	0.0	0.00	39.28	2.80	0.00	0.00	2.80
0.6*DL+0.7*EL2	-2.80	1.29	0.00	0.0	0.00	23.32	-2.80	0.00	0.00	2.80

Calculation of Base Pressure :

Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$
Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m^2

Net SBC of soil for DL, LL= 50 T/m^2 (BH28) For WL/SL case = 62.5 T/m^2
Gross SBC of soil for DL, LL= 51.8 T/m^2 For WL/SL case = 64.3 T/m^2



Load Case	Design Load/moments			Base Pressure (T/m^2)								
	V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ $Lx.(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-gh)	Remarks
D+L	34.53	0.00	0.00	2.74	0.00	0.00	4.74	4.74	4.74	4.74	3.84	O.K
DL+0.75*L+0.75*WL1	53.00	6.45	0.00	4.21	3.07	0.00	9.28	9.28	3.14	3.14	8.38	O.K
DL+0.75*L+0.75*WL2	23.60	6.45	0.00	1.87	3.07	0.00	6.95	6.95	0.80	0.80	6.05	O.K
DL+0.75*L+0.75*0.7*EL1	42.80	2.10	0.00	3.40	1.00	0.00	6.40	6.40	4.40	4.40	5.50	O.K
DL+0.75*L+0.75*0.7*EL2	30.83	2.10	0.00	2.45	1.00	0.00	5.45	5.45	3.45	3.45	4.55	O.K
0.6*DL+WL1	52.88	8.60	0.00	4.20	4.10	0.00	10.30	10.30	2.10	2.10	9.40	O.K
0.6*DL+WL2	13.68	8.60	0.00	1.09	4.10	0.00	7.19	7.19	-1.01	-1.01	6.29	-
0.6*DL+0.7*EL1	39.28	2.80	0.00	3.12	1.33	0.00	6.45	6.45	3.79	3.79	5.55	O.K
0.6*DL+0.7*EL2	23.32	2.80	0.00	1.85	1.33	0.00	5.19	5.19	2.52	2.52	4.29	O.K

Maximum Base Pressure = 9.4 T/m2 FOR LOAD CASE 0.6DL+WL1 < Allowable Net Bearing Capacity of Soil - Hence Ok

Check for Reinforcement Reinforcement Provided @ 30Kg/m3 (refer ACI 351.3R-04)

Total Quantity of Concrete = 8.81 CUM
Reinforcement Required = 264.39 kg

Longitudinal Reinf. Provided	12 dia bar 200mmc/c @0.89 kg/m	=	312.06 kg	(Refer GA dwg MD-216-49PR-EG-CV-GAD-2024 for reinf. details)
Ties Provided	12 dia bar 200mmc/c		165.54 kg	
Total Reinforcement Provided :		=	477.6 kg	> 264.39 kg - Hence Reinforcement provided is ok

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4.3 Design of Foundation for Boiler Washing Water Tank

This Underground Tank is supported on a Foundation of size 2700x2500mm^2 & thickness 250mm at a depth of 2.4m from FGL.
Top of Foundation is at RL 910.7m whereas FGL is at 913.1m.

Dead Load of Tank =		0.5	T	(Refer Tank Drawing No.1-16110200)
Self Weight of Foundation =	2.7*2.5*0.25*2.5 =	4.2	T	
Self Weight of Soil =	2.7*2.5*2.4*1.8 =	29.2	T	
Total Dead Load =		33.9	T	
Appox. Surcharge load =		2	T/m^2	
Gross Bearing Pressure =	33.9/(2.7 x 2.5) + 2 =	7.0	T/m^2	Hence Foundation Size is OK
Net SBC of soil for DL, LL=		50	T/m^2	(BH28)
Gross SBC of soil for DL, LL=	50 + 2.4*1.8 =	54.32	T/m^2	

Check for Reinforcement

Reinforcement Provided @ 30Kg/m^3 (**refer ACI 351.3R-04**)

Total Quantity of Concrete =		1.69	CUM
Reinforcement Required =		50.63	kg
Reinforcement Provided :	(Refer GA dwg MD-216-49PR-EG-CV-GAD-2024 for reinf. details)		
12 dia bar @200mm c/c =	13*(2.5+2.7)*2*0.89 =	120.328	kg

Total Weight of Reinforcement = 120.3 kg > 50.625 kg - Hence Reinforcement provided is ok

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4.4 Design of Foundation for Blowdown Tank

This Tank is supported on a Foundation of size 800x800mm^2 & thickness 800mm at a depth of 0.3m from FGL.
Top of Foundation is at RL 913.6m whereas FGL is at 913.1m. Bottom of Foundation is at RL 912.8m

Dead Load of Tank =		0.40	T	(Refer Tank Drawing No.8159-113M)
Self Weight of Foundation =	0.8*0.8*0.8*2.5 =	1.3	T	
Total Dead Load =		1.7	T	
Wind Load Intensity in Power Plant area =		0.13	T/m^2	
Wind Load on Tank of Diameter 0.58m & height 2m from Base =		0.151	T	
Moment at base due to Wind Load =	0.1508*(1+1) =	0.302	T-m	(Assuming Wind load at 1m height from Equipment Base)
Maximum Base Pressure due to Dead + Wind Load =		6.2	T/m^2	(1.7/(0.8*0.8) +0.302*6 /(0.8^2*0.8))
Appox. Surcharge load =		2	T/m^2	
Gross Bearing Pressure =	6.1953125 + 2 =	8.2	T/m^2	< 51.8T/m2 - Hence Foundation Size is OK
Net SBC of soil =		50	T/m^2	(BH28)
Gross SBC of soil =		51.8	T/m^2	

Check for Reinforcement
Reinforcement Provided @ 30Kg/m^3 (**refer ACI 351.3R-04**)

Total Quantity of Concrete =	0.51	CUM	
Reinforcement Required =	15.36	kg	

Reinforcement Provided : (Refer GA dwg MD-216-49PR-EG-CV-GAD-2024 for reinf. details)
12 dia bar @200 c/c = (5+5)*4*0.89 = 35.6 kg

Total Weight of Reinforcement = 35.6 kg > 15.36 kg - Hence Reinforcement provided is ok

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4.5 Design of Foundation for Condensate Return Tank

This horizontal Tank is supported on 250 thick Foundation raft of size 2000mmx 2000mm. The raft is enclosed by a raft beam of 300m width & 800mm depth around the periphery .
Top of Foundation is at RL 913.9m whereas FGL is at 913.4m. Bottom of Foundation is at RL 913.1m

Dead Load of Tank =		2.00	T	(Refer Tank Drawing No.8159-113M)
Weight of Foundation Raft =	(2*2*0.25)*2.5 =	2.5	T	
Weight of Foundation Beam =	0.3*(0.8-0.25)*1.7*4*2.5 =	4.7	T	
Total Dead Load =		9.2	T	
Wind Load Intensity in Power Plant area =		0.13	T/m^2	
Wind Load on Tank of Diameter 1m & length 2m =		0.260	T	
Moment at base due to Wind Load =	0.26*(1.5+0.8) =	0.598	T-m	(Assuming Wind load at 1.5m height from Equipment Base)
Moment on each side of raft =	0.598/2 =	0.299	T-m	
Maximum Base Pressure due to Dead + Wind Load =		30.0	T/m^2	(9.2/(1.7*4*0.2) +0.299*6 /(0.2^2*2))
Appox. Surcharge load =		2	T/m^2	
Gross Bearing Pressure =	30 + 2 =	32.0	T/m^2	< 51.8T/m2 - Hence Foundation Size is OK
Net SBC of soil =		50	T/m^2	(BH28)
Gross SBC of soil =		51.8	T/m^2	

Check for Reinforcement

Reinforcement Provided @ 30Kg/m^3 (**refer ACI 351.3R-04**)

Total Quantity of Concrete =	2.87	CUM
Reinforcement Required =	86.10	kg

Reinforcement Provided : (Refer GA dwg MD-216-49PR-EG-CV-GAD-2024 for reinf. details)

12 dia bar @200mm c/c =	2*11*(2+2)*0.89 =	78.32 kg
10 dia bar @200mm c/c =	2*2*(1.8*5+1.8*5)*0.61+4*11*2*0.61 =	97.6 kg

Total Weight of Reinforcement = 175.9 kg > 86.1 kg - Hence Reinforcement provided is ok

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4.6 Design of Foundation for PACU UNIT

This Unit is supported on 150 thick Foundation raft of size 2500mmx 5000mm near Transformer Room of Power Plant area. The raft is enclosed by a raft beam of 300m width & 600mm depth around the periphery .
Top of Foundation is at RL 913.4m whereas FGL is at 913.1m. Bottom of Foundation is at RL 912.8m

Dead Load of Tank =		2.50	T	(from HVAC Input)
Weight of Foundation Raft =	(2.5*5*0.15)*2.5 =	4.7	T	
Weight of Foundation Beam =	4.14*(0.6-0.15)*2.5 =	4.7	T	Base Area of Foundation Beam = 0.3*((5-0.3)+(2.5-0.3))*2 m^2 = 4.14 m^2
Total Dead Load =		11.9	T	
Appox. Surcharge load =		2	T/m^2	
Gross Bearing Pressure =	11.9/4.14 + 2 =	4.9	T/m^2	< 50.54T/m2 - Hence Foundation Size is OK

Net SBC of soil for DL, LL=		50	T/m^2	(BH28)
Gross SBC of soil for DL, LL=	50 + 0.3*1.8 =	50.54	T/m^2	

Check for Reinforcement

Reinforcement Provided @ 30Kg/m^3 (**refer ACI 351.3R-04**)

Total Quantity of Concrete =		3.74	CUM
Reinforcement Required =		112.14	kg

Reinforcement Provided :

12 dia bar @0.89kg/m =	2*(5+2.5)*8*0.89 =	106.8	kg
8 dia bar @0.39 kg/m =	2*(4.5*13+2*25)*0.39 =	84.63	kg

Total Weight of Reinforcement = 191.4 kg > 112.14 kg - Hence Reinforcement provided is ok

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4.7 Design of Foundation for BLACK START DG

This Unit is supported on 150 thick Foundation raft of size 2.838mx 6.458m near Transformer Room of Power Plant area. The raft is enclosed by a raft beam of 300m width & 450mm depth around the periphery .
Top of Foundation is at RL 913.25m whereas FGL is at 913.1m. Bottom of Foundation is at RL 912.8m

Dead Load of Tank =		12.00	T	(from HVAC Input)
Weight of Foundation Raft =	(2.838*6.458*0.15)*2.5 =	6.9	T	
Weight of Foundation Beam =	5.22*(0.45-0.15)*2.5 =	3.9	T	Base Area of Foundation Beam = 0.3*((6.458-0.3)+(2.838-0.3))*2 m^2 = 5.22 m^2
Total Dead Load =		22.8	T	
Appox. Surcharge load =		2	T/m^2	
Gross Bearing Pressure =	22.8/5.2176 + 2 =	6.4	T/m^2	< 50.54T/m2 - Hence Foundation Size is OK

Net SBC of soil for DL, LL= 50 T/m^2 (BH28)
Gross SBC of soil for DL, LL= 50 + 0.3*1.8 = 50.54 T/m^2

Check for Reinforcement

Reinforcement Provided @ 30Kg/m^3 (**refer ACI 351.3R-04**)

Total Quantity of Concrete = 4.32 CUM

Reinforcement Required = 129.46 kg

Reinforcement Provided :

12 dia bar @0.89kg/m =	2*(6.458+2.838)*6*0.89 =	99.28	kg
8 dia bar @0.39 kg/m =	2*(6.458*15+2.838*33)*0.39 =	148.61	kg

Total Weight of Reinforcement = 247.9 kg > 129.455118 kg - Hence Reinforcement provided is ok

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4.8 Design of Sump Pit

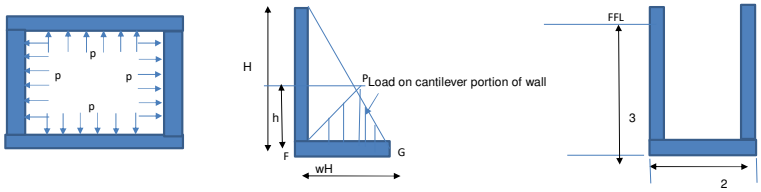
Sump Pit of 2m Width & 3m high is modelled in Staad-Pro with relevant loads. Top of concrete for Pit walls is at RL +913.6M

There is compacted structural fill below slab of Sump Pit. Sump Pit has been designed as Rectangular tank according to "Treasure of RCC Design" by Sushil Kumar (Relevant pages are attcahed for reference).

Length of Pit = 2 m Width of Pit = 2 m Height of Pit = 3 m

L/B = 1 < 2 Pit wall shall be designed as horizontal slab continuous all round (between corners) and subjected to triangular loading due to water pressure varying from zero at top to the maximum at a height h = H/4 or 1m whichever is more. Bottom portion of wall for height h is assumed to be fixed at base & designed as vertical cantilever.

Bottom portion is subjected to triangular loading (hatched portion in diagram- varying from 0 at P to wH at base)



H/4 = 0.75 m Pressure to be calculated at a height of 1 m from base

Horizontal Bending Moment for wall above P

Max. press. for horizontal bending per unit height at level of P = w(H-h) = 2 T/m^2

Horizontal Bending Moment at end of span = $P*B^2/12$ = 0.7 T-m/m height (producing tension on water face)

Horizontal Bending Moment at center of span = $P*B^2/16$ = 0.5 T-m/m height (producing tension on outer face)

Tension on each Long Wall = 1/2 * w *(H-h)*B *1 = 1.00 T/m

Tension on each Short Wall Wall = 1/2 * w *(H-h)*L *1 = 1.00 T/m

Shear at end of Wall = w(H-h) *B/2 = 2 * 2/2 = 2.00 T/m

Vertical Bending Moment for wall below P

Cantilever moment = w*H* h/2*h/3 = 1*3* 1/2* 1/3= 0.5 T-m/m

REINFORCEMENT CALCULATION OF WALL :

Wall Thickness =	200	mm
Width =	1000	mm
Clear Cover =	50	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 0.5 T-m

Myd = 0.5 T-m

ALONG X - DIRECTION :

Mu = 0.5 T-m

Φ*Kn = Mu/bd^2 deff. = 135 mm

Mu/bd^2 = 0.5X10^7/(1000X135^2) = 0.274 N/mm^2 < 6.9 N/mm^2

Pt = 0.0007 Reqd. Ast = 0.0007X1000X135/100 = 0.95 cm^2

Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal

Total Provide Ast = 5.24 cm2 at Horizontal

Total Provide Pt = 0.0039

ALONG Y - DIRECTION :

Mu = 0.5 T-m

Φ*Kn = Mu/bd^2 deff. = 145 mm

Mu/bd^2 = 0.5X10^7/(1000X145^2) = 0.238 N/mm^2 < 6.9 N/mm^2

Pt = 0.0006 Reqd. Ast = 0.0006X1000X145/100 = 0.87 cm^2

Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Vertical

Total Provide Ast = 5.24 cm2 at Vertical

Total Provide Pt = 0.0036

Minimum Reinforcement ratio for both faces = 0.0025 (Cl. No. 11.6, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

Mxd = 0.7 T-M

Myd = 0.5 T-M

ALONG X - DIRECTION :

Mu = 0.666666667 T-m

Φ*Kn = Mu/bd^2 deff. = 135 mm

Mu/bd^2 = 0.666666666666667X10^7/(1000X135^2) = 0.366 N/mm^2 < 6.9 N/mm^2

Pt = 0.0010 Reqd. Ast = 0.001X1000X135/100 = 1.35 cm^2

Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal

Total Provide Ast = 5.24 cm2) at Horizontal

Total Provide Pt = 0.0039

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	01-12-2020	07-01-2021	26-10-2023	Date	01-12-2020	07-01-2021	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

ALONG Y - DIRECTION :

Mu = 0.5 T-m

$\Phi Kn = Mu/bd^2$ deff. = 145 mm

Mu/bd^2 = 0.5X10^7/(X145^2) = 0.238 N/mm^2 < 6.9 N/mm^2

Pt = 0.0006 Reqd. Ast = 0.0006XX145/100 = 0.87 cm^2

Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Vertical

Total Provide Ast = 5.24 cm2) at Vertical

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

Vu = 2 T

Provided Pt = 0.0036

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0036X2X10^4X145/0.5/10^7)X1000X145/10^4

= 14.24 T

[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0036)X1000X145/10^4

= 14.613 T

[c] = 0.29*λ*fc'^0.5*b*d.eff

= 0.29X1X5.92X1000X145/10^4

= 24.877 T

HENCE, Vc = 14.24 T

ΦVc = 0.75X14.24 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 10.68 T **> 2T , Hence O.K.**

Bending Moment & Shear Force in Raft due to Water Pressure

Bending Moment in Raft : wH*B^2/16 = 1*3*2^2/16 = 0.75 T-m/m width (bottom tension)

wH*B^2/12 = 1*3*2^2/12 = 1.000 T-m/m width (Top tension)

Shear force at face of wall = wH *B/2 = 1* 3 * 2/2 = 3.00 T/m

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	C01	D01
Date	01-12-2020	07-01-2021	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	01-12-2020	07-01-2021	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

REINFORCEMENT CALCULATION OF WALL RAFT :

Slab Thickness = 200 mm
Width = 1000 mm
Clear Cover = 50 mm
Compressive Strength of Concrete , fc' = 35 N/mm^2
Yield Strength of Main Reinforcement , fy = 420 N/mm^2
Yield Strength of Shear Reinforcement , fyt = 420 N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ = 0.9
Strength Reduction Factor of Shear Reinforcement , Φt = 0.75

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 0.75 T-m
Myd = 0.75 T-m

ALONG X - DIRECTION :

Mu = 0.75 T-m
Φ*Kn = Mu/bd^2 deff. = 145 mm
Mu/bd^2 = 0.75X10^7/(1000X145^2) = 0.357 N/mm^2 < 6.9 N/mm^2
Pt = 0.0010 Reqd. Ast = 0.001X1000X145/100 = 1.45 cm^2
Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal
Total Provide Ast = 5.24 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0036

ALONG Y - DIRECTION :

Mu = 0.75 T-m
Φ*Kn = Mu/bd^2 deff. = 135 mm
Mu/bd^2 = 0.75X10^7/(1000X135^2) = 0.412 N/mm^2 < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X135/100 = 1.49 cm^2
Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal
Total Provide Ast = 5.24 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0039
Minimum Reinforcement ratio for both faces = 0.0018 (Cl. No. Table 8.6.1.1, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

-0.036

Mxd = 1 T-M
Myd = 1 T-M

ALONG X - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2 deff. = 145 mm
Mu/bd^2 = 1X10^7/(1000X145^2) = 0.476 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X145/100 = 1.89 cm^2
Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal
Total Provide Ast = 5.24 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0036

ALONG Y - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2 deff. = 135 mm
Mu/bd^2 = 1X10^7/(1000X135^2) = 0.549 N/mm^2 < 6.9 N/mm^2
Pt = 0.0015 Reqd. Ast = 0.0015X1000X135/100 = 2.03 cm^2
Provided **10** mm @ **150** C/C (Provide Ast = 5.24 cm2) at Horizontal
Total Provide Ast = 5.24 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0039

SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

Vu = 3 T

Provided Pt = 0.0036
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0036X3X10^4X145/1/10^7)X1000X145/10^4
= 14.111 T
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0036)X1000X135/10^4
= 13.605 T
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X135/10^4
= 23.161 T
HENCE, Vc = 13.605 T
Φt*Vc = 0.75X13.605 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 10.204 T **> 3T , Hence O.K.**

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Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

ATTACHMENT 1 STAAD INPUT & OUTPUT FILE OF BOILER FOUNDATION

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    DEC  1, 2020                *
*           Time=    14:33:47                    *
*                                                *
*      USER ID:                                  *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: E:\SHARE\MAADEN\DG POWER PLANT\MISC. FOUNDATION\Boiler Foundation-R.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 0; 14 0 -4 -0.25; 16 0 -4 1.2; 20 1.25 -4 1.2; 21 0.75 -4 1.2
     9. 22 0.75 -4 0.95; 23 1.25 -4 0.95; 24 0.5 -4 1.2; 25 0.5 -4 0.95
    10. 26 0.25 -4 1.2; 27 0.25 -4 0.95; 28 0 -4 0.95; 29 0.75 -4 0.75
    11. 30 1.25 -4 0.75; 31 0.5 -4 0.75; 32 0.25 -4 0.75; 33 0 -4 0.75
    12. 34 0.75 -4 0.25; 35 1.25 -4 0.25; 36 0.5 -4 0.25; 37 0.25 -4 0.25
    13. 38 0 -4 0.25; 39 0.75 -4 0; 40 1.25 -4 0; 41 0.5 -4 0; 42 0.25 -4 0
    14. 43 0.75 -4 -0.25; 44 1.25 -4 -0.25; 45 0.5 -4 -0.25; 46 0.25 -4 -0.25
    15. 72 0.75 -4 -0.5; 73 1.25 -4 -0.5; 74 0.5 -4 -0.5; 75 0.25 -4 -0.5
    16. 76 0 -4 -0.5; 77 0.75 -4 -0.75; 78 1.25 -4 -0.75; 79 0.5 -4 -0.75
    17. 80 0.25 -4 -0.75; 81 0 -4 -0.75; 82 0.75 -4 -1; 83 1.25 -4 -1; 84 0.5 -4 -1.
    18. 85 0.25 -4 -1; 86 0 -4 -1; 90 -0.25 -4 1.2; 91 -0.25 -4 0.95; 92 -0.5 -4 1.2
    19. 93 -0.5 -4 0.95; 97 -0.25 -4 0.75; 98 -0.5 -4 0.75; 101 -0.25 -4 0.25
    20. 102 -0.5 -4 0.25; 105 -0.25 -4 0; 106 -0.5 -4 0; 109 -0.25 -4 -0.25
    21. 110 -0.5 -4 -0.25; 133 -0.25 -4 -0.5; 134 -0.5 -4 -0.5; 137 -0.25 -4 -0.75
    22. 138 -0.5 -4 -0.75; 141 -0.25 -4 -1; 142 -0.5 -4 -1; 145 0 -2 -0.75
    23. 146 0 -2 0.95; 147 0.75 -4 -1.25; 148 1.25 -4 -1.25; 149 0.5 -4 -1.25
    24. 150 0.25 -4 -1.25; 151 0 -4 -1.25; 152 -0.25 -4 -1.25; 153 -0.5 -4 -1.25
    25. 156 1.25 -4 1.45; 157 0.75 -4 1.45; 158 0.5 -4 1.45; 159 0.25 -4 1.45
    26. 160 0 -4 1.45; 161 -0.25 -4 1.45; 162 -0.5 -4 1.45; 181 1.7 -4 1.2
    27. 182 1.7 -4 0.95; 183 1.7 -4 0.75; 184 1.7 -4 0.25; 185 1.7 -4 0
    28. 186 1.7 -4 -0.25; 192 1.7 -4 -0.5; 193 1.7 -4 -0.75; 194 1.7 -4 -1.
    29. 195 1.7 -4 -1.25; 196 1.7 -4 1.45; 220 1.7 -2 -0.75; 222 1.7 -2 0.95
    30. 361 1.95 -4 1.2; 362 1.95 -4 0.95; 363 1.95 -4 0.75; 364 1.95 -4 0.25
    31. 365 1.95 -4 0; 366 1.95 -4 -0.25; 372 1.95 -4 -0.5; 373 1.95 -4 -0.75
    32. 374 1.95 -4 -1; 375 1.95 -4 -1.25; 376 1.95 -4 1.45; 419 2.2 -4 1.2
    33. 420 2.2 -4 0.95; 421 2.2 -4 0.75; 422 2.2 -4 0.25; 423 2.2 -4 0
    34. 424 2.2 -4 -0.25; 430 2.2 -4 -0.5; 431 2.2 -4 -0.75; 432 2.2 -4 -1.
    35. 433 2.2 -4 -1.25; 434 2.2 -4 1.45; 507 1.5 -4 1.2; 508 1.5 -4 0.95
    36. 509 1.5 -4 0.75; 510 1.5 -4 0.25; 511 1.5 -4 0; 512 1.5 -4 -0.25
    37. 513 1.5 -4 -0.5; 514 1.5 -4 -0.75; 515 1.5 -4 -1; 516 1.5 -4 -1.25
    38. 517 1.5 -4 1.45; 631 1 -4 1.2; 632 1 -4 0.95; 633 1 -4 0.75; 634 1 -4 0.25
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39. 635 1 -4 0; 636 1 -4 -0.25; 637 1 -4 -0.5; 638 1 -4 -0.75; 639 1 -4 -1.
40. 640 1 -4 -1.25; 641 1 -4 1.45; 646 1.25 -4 0.5; 647 0.75 -4 0.5
41. 648 0.5 -4 0.5; 649 0.25 -4 0.5; 650 0 -4 0.5; 651 -0.25 -4 0.5
42. 652 -0.5 -4 0.5; 653 1.95 -4 0.5; 654 1.7 -4 0.5; 655 2.2 -4 0.5
43. 656 1.5 -4 0.5; 657 1 -4 0.5
44. MEMBER INCIDENCES
45. 133 81 145; 134 28 146; 201 193 220; 202 182 222
46. ELEMENT INCIDENCES SHELL
47. 26 21 24 25 22; 27 24 26 27 25; 28 26 16 28 27; 30 22 25 31 29; 31 25 27 32 31
48. 32 27 28 33 32; 38 34 36 41 39; 39 36 37 42 41; 40 37 38 13 42; 42 39 41 45 43
49. 43 41 42 46 45; 44 42 13 14 46; 66 43 45 74 72; 67 45 46 75 74; 68 46 14 76 75
50. 70 72 74 79 77; 71 74 75 80 79; 72 75 76 81 80; 74 77 79 84 82; 75 79 80 85 84
51. 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 85 28 91 97 33; 86 91 93 98 97
52. 93 38 101 105 13; 94 101 102 106 105; 97 13 105 109 14; 98 105 106 110 109
53. 121 14 109 133 76; 122 109 110 134 133; 125 76 133 137 81; 126 133 134 138 137
54. 129 81 137 141 86; 130 137 138 142 141; 136 82 84 149 147; 137 84 85 150 149
55. 138 85 86 151 150; 139 86 141 152 151; 140 141 142 153 152; 144 157 158 24 21
56. 145 158 159 26 24; 146 159 160 16 26; 147 160 161 90 16; 148 161 162 92 90
57. 335 361 181 182 362; 336 362 182 183 363; 338 364 184 185 365
58. 339 365 185 186 366; 345 366 186 192 372; 346 372 192 193 373
59. 347 373 193 194 374; 348 374 194 195 375; 349 376 196 181 361
60. 375 419 361 362 420; 376 420 362 363 421; 378 422 364 365 423
61. 379 423 365 366 424; 385 424 366 372 430; 386 430 372 373 431
62. 387 431 373 374 432; 388 432 374 375 433; 389 434 376 361 419
63. 458 507 20 23 508; 459 508 23 30 509; 461 510 35 40 511; 462 511 40 44 512
64. 464 512 44 73 513; 465 513 73 78 514; 466 514 78 83 515; 467 515 83 148 516
65. 468 517 156 20 507; 587 181 507 508 182; 588 182 508 509 183
66. 590 184 510 511 185; 591 185 511 512 186; 593 186 512 513 192
67. 594 192 513 514 193; 595 193 514 515 194; 596 194 515 516 195
68. 597 196 517 507 181; 603 631 21 22 632; 604 632 22 29 633; 606 634 34 39 635
69. 607 635 39 43 636; 608 636 43 72 637; 609 637 72 77 638; 610 638 77 82 639
70. 611 639 82 147 640; 612 641 157 21 631; 618 647 648 36 34; 619 648 649 37 36
71. 620 649 650 38 37; 621 650 651 101 38; 622 651 652 102 101
72. 623 653 654 184 364; 624 655 653 364 422; 625 656 646 35 510
73. 626 654 656 510 184; 627 657 647 34 634; 628 20 631 632 23; 629 23 632 633 30
74. 631 35 634 635 40; 632 40 635 636 44; 633 44 636 637 73; 634 73 637 638 78
75. 635 78 638 639 83; 636 83 639 640 148; 637 156 641 631 20; 642 646 657 634 35
76. 643 29 31 648 647; 644 31 32 649 648; 645 32 33 650 649; 646 33 97 651 650
77. 647 97 98 652 651; 648 363 183 654 653; 649 421 363 653 655
78. 650 509 30 646 656; 651 183 509 656 654; 652 633 29 647 657
79. 653 30 633 657 646
80. START GROUP DEFINITION
81. ELEMENT
82. _MOMENT 26 27 30 31 38 TO 40 42 TO 44 66 TO 68 70 71 74 75 82 86 93 94 97 -
83. 98 121 122 126 130 136 TO 140 144 TO 148 338 339 345 348 349 375 376 378 -
84. 379 385 TO 389 458 459 461 462 464 TO 468 590 591 593 596 597 603 604 606 -
85. 607 TO 612 618 TO 629 631 TO 637 642 TO 653
86. _SHEAR 26 30 38 TO 40 42 TO 44 66 70 74 93 94 97 98 136 144 338 339 378 379 -
87. 461 462 590 591 603 604 606 TO 612 618 TO 629 631 TO 637 642 TO 653
88. END GROUP DEFINITION
89. ELEMENT PROPERTY
90. 26 TO 28 30 TO 32 38 TO 40 42 TO 44 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 -
91. 93 94 97 98 121 122 125 126 129 130 136 TO 140 144 TO 148 335 336 338 339 -
92. 345 TO 349 375 376 378 379 385 TO 389 458 459 461 462 464 TO 468 587 588 -
93. 590 591 593 TO 597 603 604 606 TO 612 618 TO 629 631 TO 637 642 TO 652 -
94. 653 THICKNESS 0.25
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95. SUPPORTS
96. 13 14 16 20 TO 46 72 TO 86 90 TO 93 97 98 101 102 105 106 109 110 133 134 -
97. 137 138 141 142 147 TO 153 156 TO 162 181 TO 186 192 TO 196 361 TO 366 372 -
98. 373 TO 376 419 TO 424 430 TO 434 507 TO 517 631 TO 641 646 TO 656 -
99. 657 ELASTIC MAT DIRECT Y SUBGRADE 7080 PRINT
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JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
20	0.062500
21	0.062500
22	0.056250
23	0.056250
24	0.062500
25	0.056250
26	0.062500
27	0.056250
28	0.056250
29	0.056250
30	0.056250
31	0.056250
32	0.056250
33	0.056250
34	0.062500
35	0.062500
36	0.062500
37	0.062500
38	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062500
45	0.062500
46	0.062500
72	0.062500
73	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062500
80	0.062500
81	0.062500
82	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.062500
90	0.062500
91	0.056250

92	0.031250
93	0.028125
97	0.056250
98	0.028125
101	0.062500
102	0.031250
105	0.062500
106	0.031250
109	0.062500
110	0.031250
133	0.062500
134	0.031250
137	0.062500
138	0.031250
141	0.062500
142	0.031250
147	0.031250
148	0.031250
149	0.031250
150	0.031250
151	0.031250
152	0.031250
153	0.015625
156	0.031250
157	0.031250
158	0.031250
159	0.031250
160	0.031250
161	0.031250
162	0.015625
181	0.056250
182	0.050625
183	0.050625
184	0.056250
185	0.056250
186	0.056250
192	0.056250
193	0.056250
194	0.056250
195	0.028125
196	0.028125
361	0.062500
362	0.056250
363	0.056250
364	0.062500
365	0.062500
366	0.062500
372	0.062500
373	0.062500
374	0.062500
375	0.031250
376	0.031250
419	0.031250
420	0.028125
421	0.028125
422	0.031250

423	0.031250
424	0.031250
430	0.031250
431	0.031250
432	0.031250
433	0.015625
434	0.015625
507	0.056250
508	0.050625
509	0.050625
510	0.056250
511	0.056250
512	0.056250
513	0.056250
514	0.056250
515	0.056250
516	0.028125
517	0.028125
631	0.062500
632	0.056250
633	0.056250
634	0.062500
635	0.062500
636	0.062500
637	0.062500
638	0.062500
639	0.062500
640	0.031250
641	0.031250
646	0.062500
647	0.062500
648	0.062500
649	0.062500
650	0.062500
651	0.062500
652	0.031250
653	0.062500
654	0.056250
655	0.031250
656	0.056250
657	0.062500
TOTAL	7.290000

```
100. DEFINE MATERIAL START
101. ISOTROPIC CONCRETE
102. E 2.21467E+006
103. POISSON 0.17
104. DENSITY 2.40262
105. ALPHA 1E-005
106. DAMP 0.05
107. TYPE CONCRETE
108. STRENGTH FCU 2812.28
109. END DEFINE MATERIAL
110. MEMBER PROPERTY AMERICAN
111. 133 134 201 202 PRIS YD 0.7 ZD 0.7
112. CONSTANTS
113. MATERIAL CONCRETE ALL
```

```
114. ************************************************************************
115. LOAD 1 DL
116. SELFWEIGHT Y -1
117. ELEMENT LOAD
118. 26 TO 28 30 TO 32 38 TO 40 42 TO 44 66 TO 68 70 TO 72 74 TO 76 81 82 85 86 -
119. 93 94 97 98 121 122 125 126 129 130 136 TO 140 144 TO 148 335 336 338 339 -
120. 345 TO 349 375 376 378 379 385 TO 389 458 459 461 462 464 TO 468 587 588 -
121. 590 591 593 TO 597 603 604 606 TO 612 618 TO 629 631 TO 637 642 TO 652 -
122. 653 PR GY -3.2
123. JOINT LOAD
124. 145 146 220 222 FY -3.3
125. ************************************************************************8
126. LOAD 2 LL
127. JOINT LOAD
128. 145 146 220 222 FY -1.25
129. ************************************************************************8
130. LOAD 3 WIND IN X DIR (WL+X)
131. JOINT LOAD
132. 145 146 220 222 FX 0.5
133. 220 222 FY -1.5
134. 145 146 FY 1.5
135. ************************************************************************8
136. LOAD 4 WIND IN Z DIR (WL+Z)
137. JOINT LOAD
138. 145 146 220 222 FZ 0.5
139. 146 222 FY -1.5
140. 145 220 FY 1.5
141. ************************************************************************
142. LOAD 5 SEISMIC IN X DIR (SL+X)
143. JOINT LOAD
144. 145 146 220 222 FX 1
145. 220 222 FY -2.5
146. 145 146 FY 2.5
147. ************************************************************************
148. LOAD 6 SEISMIC IN Z DIR (SL+Z)
149. JOINT LOAD
150. 145 146 220 222 FZ 1
151. 146 222 FY -2.5
152. 145 220 FY 2.5
153. ************************************************************************
154. *LOAD COMBINATIONS FOR STABILITY CHECK
155. ************************************************************************8
156. LOAD COMB 100 DL+LL
157. 1 1.0 2 1.0
158. LOAD COMB 101 DL+0.75LL+0.75*WL(+X)
159. 1 0.75 2 0.75 3 0.75
160. LOAD COMB 102 DL+0.75LL+0.75*WL(-X)
161. 1 0.75 2 0.75 3 -0.75
162. LOAD COMB 103 DL+0.75LL+0.75*WL(+Z)
163. 1 0.75 2 0.75 4 0.75
164. LOAD COMB 104 DL+0.75LL+0.75*WL(-Z)
165. 1 0.75 2 0.75 4 -0.75
166. LOAD COMB 105 DL+0.75LL+0.75*0.7*SL(+X)
167. 1 0.75 2 0.75 5 0.525
168. LOAD COMB 106 DL+0.75LL+0.75*0.7*SL(-X)
169. 1 0.75 2 0.75 5 -0.525
```

```
LOAD COMB 107 DL+0.75LL+0.75*0.7*SL(+Z)
1 0.75 2 0.75 6 0.525
LOAD COMB 108 DL+0.75LL+0.75*0.7*SL(-Z)
1 0.75 2 0.75 6 -0.525
LOAD COMB 109 0.6*DL+WL(+X)
1 0.6 3 1.0
LOAD COMB 110 0.6*DL+WL(-X)
1 0.6 3 -1.0
LOAD COMB 111 0.6*DL+WL(+Z)
1 0.6 4 1.0
LOAD COMB 112 0.6*DL+WL(-Z)
1 0.6 4 -1.0
LOAD COMB 113 0.6*DL+0.7*SL(+X)
1 0.6 5 0.7
LOAD COMB 114 0.6*DL+0.7*SL(-X)
1 0.6 5 -0.7
LOAD COMB 115 0.6*DL+0.7*SL(+Z)
1 0.6 6 0.7
LOAD COMB 116 0.6*DL+0.7*SL(-Z)
1 0.6 6 -0.7
**************************************************************************
*LOAD COMBINATIONS FOR FOUNDATION DESIGN
LOAD COMB 200 1.4D
1 1.4
LOAD COMB 201 1.4D+1.7(LL)
1 1.4 2 1.7
LOAD COMB 202 1.2D+1.0(LL)+1.6WL(+X)
1 1.2 2 1.0 3 1.6
LOAD COMB 203 1.2D+1.0(LL)+1.6WL(-X)
1 1.2 2 1.0 3 -1.6
LOAD COMB 204 1.2D+1.0(LL)+1.6WL(+Z)
1 1.2 2 1.0 4 1.6
LOAD COMB 205 1.2D+1.0(LL)+1.6WL(-Z)
1 1.2 2 1.0 4 -1.6
LOAD COMB 206 0.9*D+1.6WL(+X)
1 0.9 3 1.6
LOAD COMB 207 0.9*D+1.6WL(-X)
1 0.9 3 -1.6
LOAD COMB 208 0.9*D+1.6WL(+Z)
1 0.9 4 1.6
LOAD COMB 209 0.9*D+1.6WL(-Z)
1 0.9 4 -1.6
LOAD COMB 210 1.2D+1.0(LL)+1.0SL(+X)
1 1.2 2 1.0 5 1.0
LOAD COMB 211 1.2D+1.0(LL)+1.0SL(-X)
1 1.2 2 1.0 5 -1.0
LOAD COMB 212 1.2D+1.0(LL)+1.0SL(+Z)
1 1.2 2 1.0 6 1.0
LOAD COMB 213 1.2D+1.0(LL)+1.0SL(-Z)
1 1.2 2 1.0 6 -1.0
LOAD COMB 214 0.9*D+1.0SL(+X)
1 0.9 5 1.0
LOAD COMB 215 0.9*D+1.0SL(-X)
1 0.9 5 -1.0
LOAD COMB 216 0.9*D+1.0SL(+Z)
1 0.9 6 1.0
```

```
226. LOAD COMB 217 0.9*D+1.0SL(-Z)
227. 1 0.9 6 -1.0
228. ********************************************************************
229. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**148**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**121**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**144**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 135/ 20/ 63 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 456
TOTAL LOAD COMBINATION CASES = 35 SO FAR.
SIZE OF STIFFNESS MATRIX = 29 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 13.0/ 271796.9 MB

```
230. LOAD LIST 200 TO 217
231. START CONCRETE DESIGN
```

```
232. CODE ACI 1999
233. TRACK 0 MEMB 133 134
234. MAXMAIN 25 MEMB 133 134
235. CLB 0.075 MEMB 133 134
236. CLS 0.075 MEMB 133 134
237. END CONCRETE DESIGN
238. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= DEC  1,2020   TIME= 14:33:48 ****

       ************************************************************
       *   For technical assistance on STAAD.Pro, please visit    *
       *   http://selectservices.bentley.com/en-US/               *
       *                                                          *
       *   Details about additional assistance from               *
       *   Bentley and Partners can be found at program menu      *
       *   Help->Technical Support                                *
       *                                                          *
       *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
       *                http://www.bentley.com                    *
       ************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	01-12-2020	07-01-2021	26-10-2023				Date	01-12-2020	07-01-2021	26-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD			
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 EQUIPMENT DATA SHEET FOR BOILER & OTHER EQUIPMENTS

MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

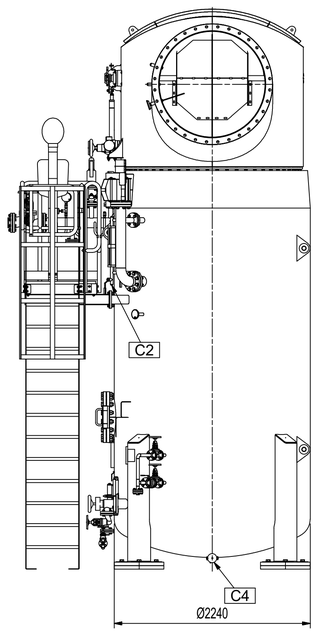
Title: **Boiler GA Drawing**
49PR POWER GENERATION
Drawing

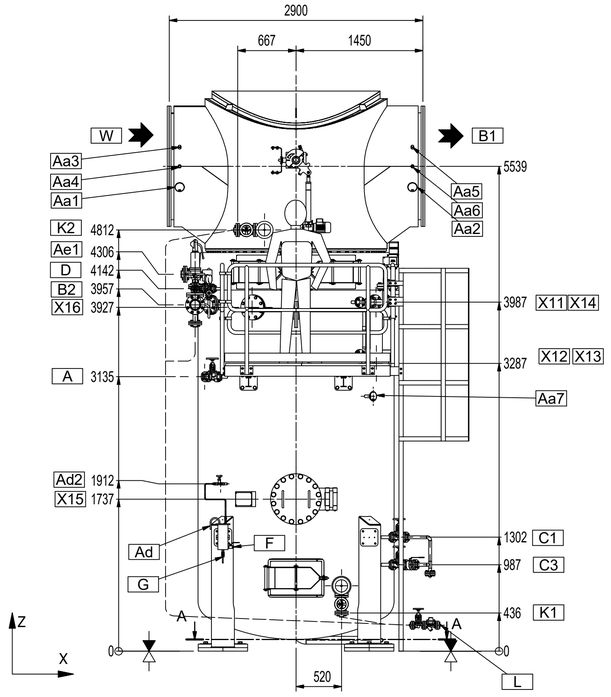
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Revision:
Revision Date:
Status:

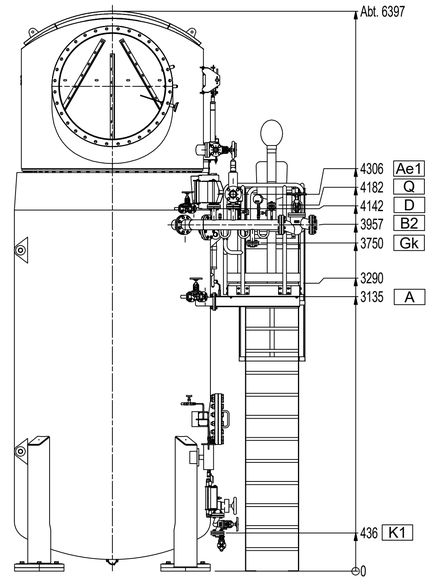
Outotec Document Number:							
L&T Document Number:							

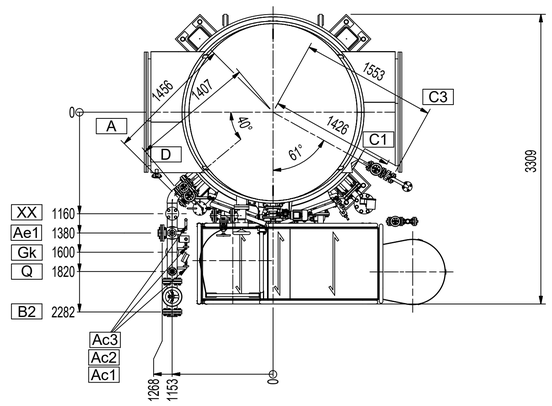
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	Wartsila Doc no: DBAF490847 a
	Original Doc no: 8159-69M
	Wartsila Supplier:

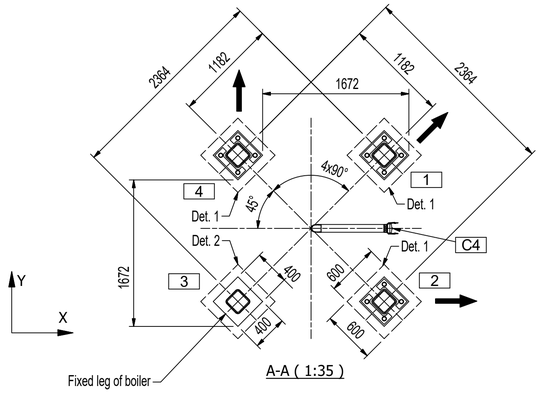
REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**





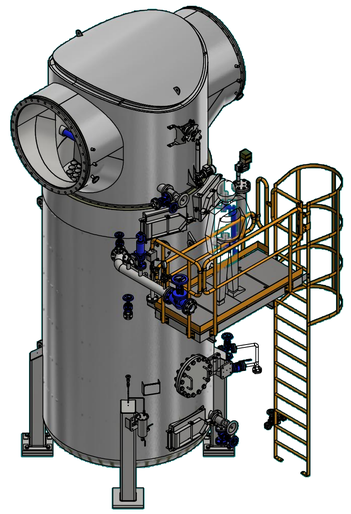




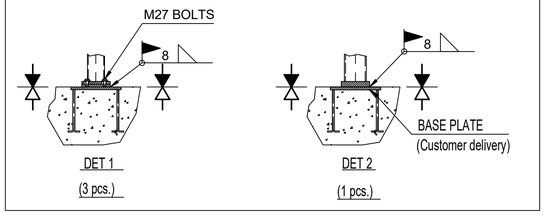


A-A (1:35)

Thermal expansion has been taken into consideration in leg construction



FOUNDATIONS FOR BOILER



ASSEMBLY TOLERANCES FOR FOUNDATIONS

THE FOLLOWING GENERAL TOLERANCES ARE TO BE FOLLOWED DURING ASSEMBLY OF BASE PLATES OR ANCHOR BOLTS TO THE CONCRETE BED:

CENTER LINE DISTANCE INSIDE ONE ANCHOR BOLT GROUP	±2
DIAGONAL DIMENSION INSIDE ONE ANCHOR BOLT GROUP	±3
ELEVATION POSITION FOR ANCHOR BOLTS	±10
CENTER LINE DISTANCE BETWEEN ANCHOR BOLT GROUPS	±5
STRAIGHTNESS OF BASE PLATES	±2
CENTER LINE DISTANCE BETWEEN BASE PLATES	±5
ELEVATION POSITION FOR BASE PLATES	±5

GENERAL TOLERANCE FOR TERMINAL POINTS: ±20

THE SLIDING SUPPORTS AND FIXED POINTS OF THE PIPELINES TO BE DESIGNED SO THAT THE MASS FORCES AND THERMAL EXPANSIONS ARE COMPENSATED.

WEIGHT INFORMATION TABLE

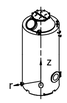
DRY WEIGHT:	13180 kg
MAX. WEIGHT:	18300 kg
OPERATING WEIGHT:	17800 kg

DELIVERY LIMITS ARE SHOWN BY — ALFA LAVAL AALBORG OY — OTHERS

Fax.	Flat.	Fvert.	Mtors.	Mlat.	Mvert.	r	z	Pos	Connection	Size	PN	Note
Max. Loads (kN)			Max. Moments (kNm)			Thermal Movements (mm), T=Design						
x	x	x	x	x	x	x	x	Q	AIR FOR UNDERPRESSURE PROTECTION	DN15	PN15	
x	x	x	x	x	x	x	x	XX	SPARE	DN40	PN40	
x	x	x	x	x	x	x	x	X16	OBSERVATION HATCH	DN150	PN16	
x	x	x	x	x	x	x	x	X15	OBSERVATION HATCH	DN500	PN16	
x	x	x	x	x	x	x	x	X14	SENSOR TUBE CONNECTION	DN50	PN40	
x	x	x	x	x	x	x	x	X13	SENSOR TUBE CONNECTION	DN100	PN40	
x	x	x	x	x	x	x	x	X12	LEVEL GAUGE CONNECTION	DN25	PN40	
x	x	x	x	x	x	x	x	X11	LEVEL GAUGE CONNECTION	DN25	PN40	
5	5	5	5	5	5	4,3	10,6	W	EXHAUST GAS IN	DN1200	PN2.5	WDAAA205343
x	x	x	x	x	x	x	x	L	CONDENSATE/DRAIN	DN15	PN40	
0,2	0,2	0,2	0,3	0,3	0,3	4,7	10,6	K2	SOOTBLOWER STEAM IN	DN40	PN40	
0,2	0,2	0,2	0,3	0,3	0,3	4,6	-1,4	K1	SOOTBLOWER STEAM IN	DN40	PN40	
0,2	0,2	0,2	0,05	0,05	0,05	4,5	5,9	Gk	STEAM TO SOOTBLOWERS	DN40	PN40	
x	x	x	x	x	x	x	x	G	COOLING WATER OUT	G½" (F)		
x	x	x	x	x	x	x	x	F	COOLING WATER IN	G½" (F)		
0,2	0,2	0,2	0,1	0,1	0,1	2,8	6,8	D	VENTING	DN20	PN40	
x	x	x	x	x	x	x	x	C4	BOILER WASHING WATER OUT	R3"		Camlock fitting (M)
x	x	x	x	x	x	x	x	C3	BOTTOM BLOW OUT	DN25	PN40	
x	x	x	x	x	x	x	x	C2	LEVEL GAUGE DRAIN	Ø12		
x	x	x	x	x	x	x	x	C1	SCUM BLOW OUT	DN25	PN40	
0,4	0,4	0,4	0,1	0,1	0,1	6	6,4	B2	MAIN STEAM OUT	DN100	PN16	
5	5	5	5	5	5	4,3	10,6	B1	EXHAUST GAS OUT	DN1200	PN2.5	WDAAA205343
0,2	0,2	0,2	0,05	0,05	0,05	4,1	7,2	Ae1	SAFETY VALVE BLOW-OUT	DN65	PN16	TO SAFE PLACE
x	x	x	x	x	x	x	x	Ad2	WATER TO SAMPLE COOLER	G½" (M)		
x	x	x	x	x	x	x	x	Ad	SAMPLE	Ø8mm		
x	x	x	x	x	x	x	x	Ac3	SPARE	G½" (ML)		
x	x	x	x	x	x	x	x	Ac2	PRESSURE SWITCH	G½" (ML)		
x	x	x	x	x	x	x	x	Ac1	LOCAL PRESSURE INDICATION	G½" (ML)		
x	x	x	x	x	x	x	x	Aa7	TEMP BOILER WATER	G½" (F)		
x	x	x	x	x	x	x	x	Aa6	SPARE	G½" (F)		
x	x	x	x	x	x	x	x	Aa5	SPARE	G½" (F)		
x	x	x	x	x	x	x	x	Aa4	SPARE	G½" (F)		
x	x	x	x	x	x	x	x	Aa3	SPARE	G½" (F)		
x	x	x	x	x	x	x	x	Aa2	LOCAL TEMPERATURE INDICATION	G½" (F)		
x	x	x	x	x	x	x	x	Aa1	LOCAL TEMPERATURE INDICATION	G½" (F)		
0,2	0,2	0,2	0,05	0,05	0,05	2,8	4,6	A	FEED WATER IN	DN25	PN40	

ALLOWABLE FORCES AND MOMENTS

- Not to be subjected to any fluctuating loads.
- All loads are only static (+-).
- Forces given in the table are not allowed to be exceeded.
- Other pipelines to be designed so that the mass forces and thermal expansions are compensated.



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Power Plants

WÄRTSILÄ

Product WHR-Steam system | ACA | ASLY.DRG

Main dimensions

Exhaust gas boiler HU-2A

RCC011, RCC021, RCC051

8159 Mansourah - Massarah Gold Mine Proj. no. *8159*

MADE	7.4.2020	FIRAJSNB	Units: mm/kg
CHKD	Ref. to:	FIRA:QMS (2018)	Weight 11650 kg
APPD	Ref. to:	FIRA:QMS (2018)	Customer Document: -
			Page No. 1 /1
			Scale: 1:35
			Size: A1
			Doc. ID: 8159-69M
			Rev. -

Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **Main dimensions- AUXILIARY BOILER BH-1500L**
49PR POWER GENERATION
Dimensional drawing

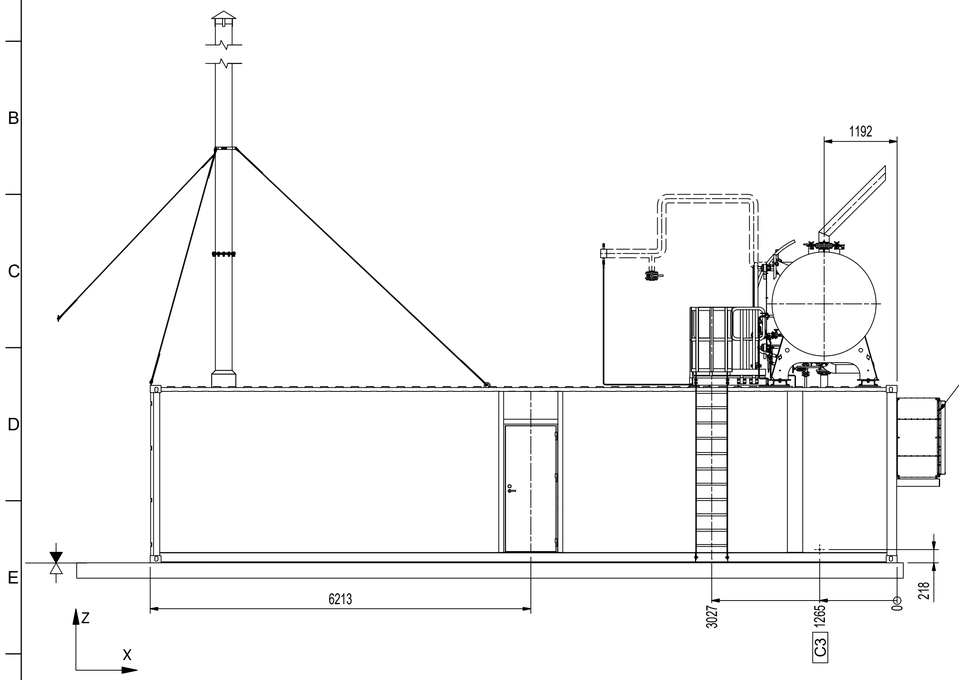
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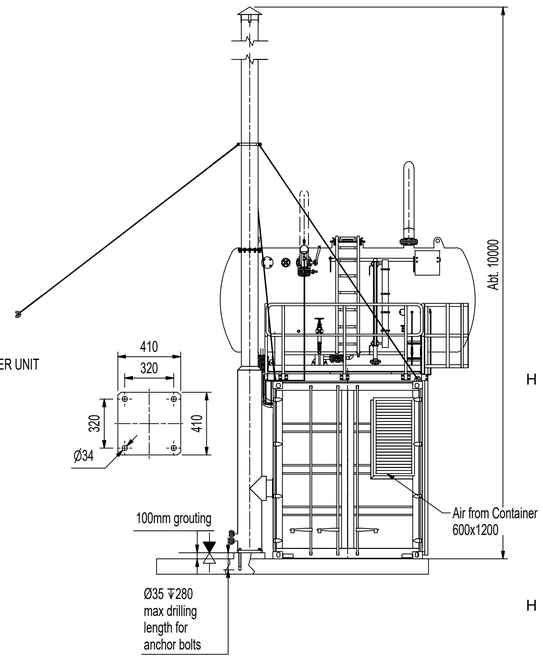
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L&T Document Number:							

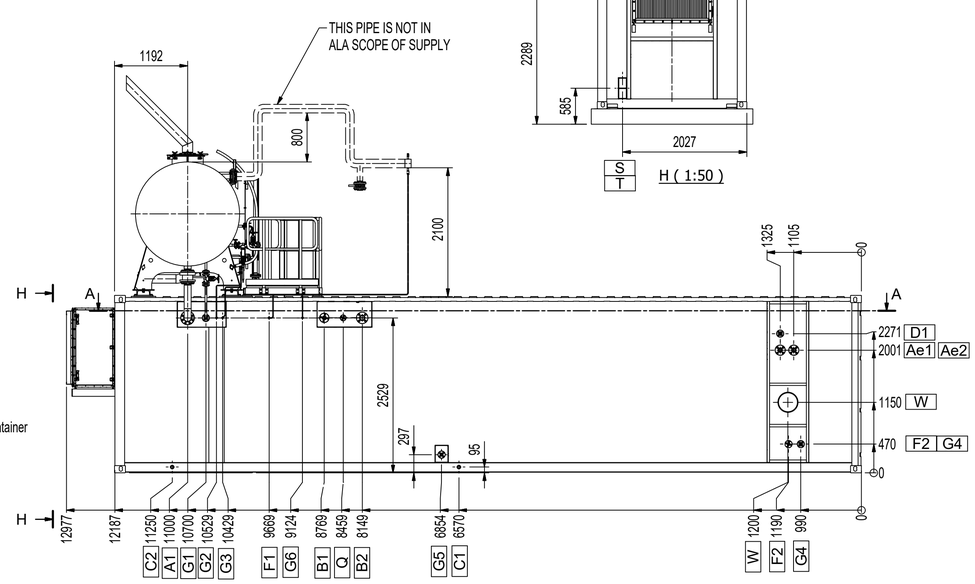
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	Wartsila Doc no: DBAF490878 a
	Original Doc no: 8159-94M
	Wartsila Supplier:

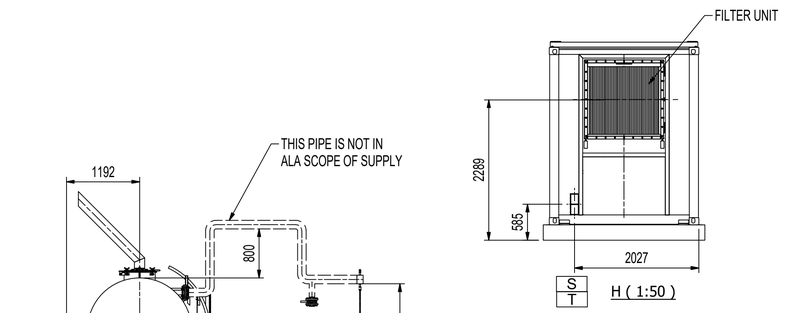
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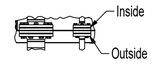
Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer



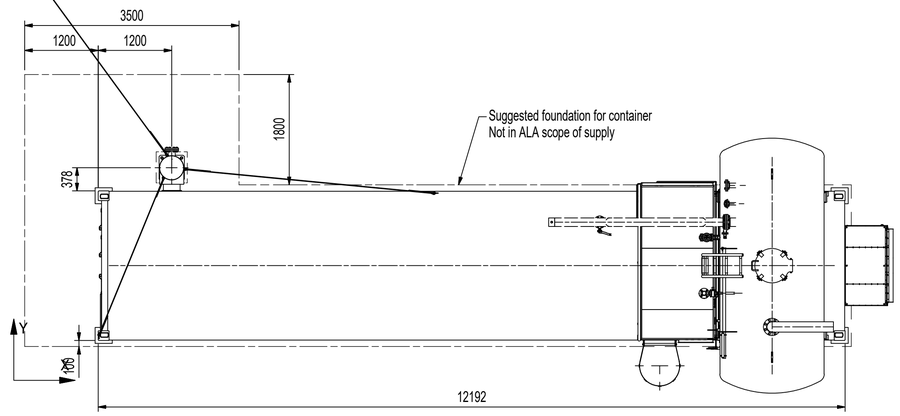


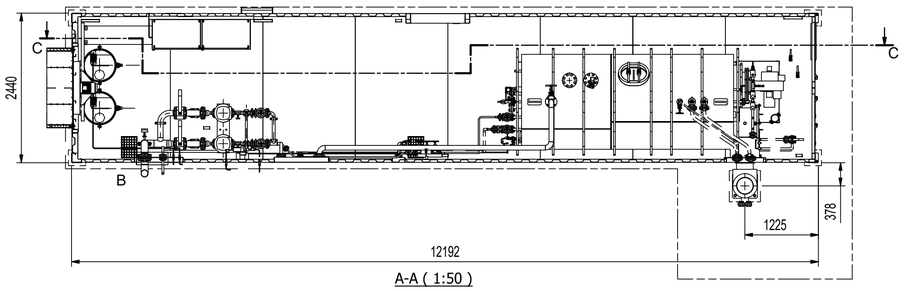




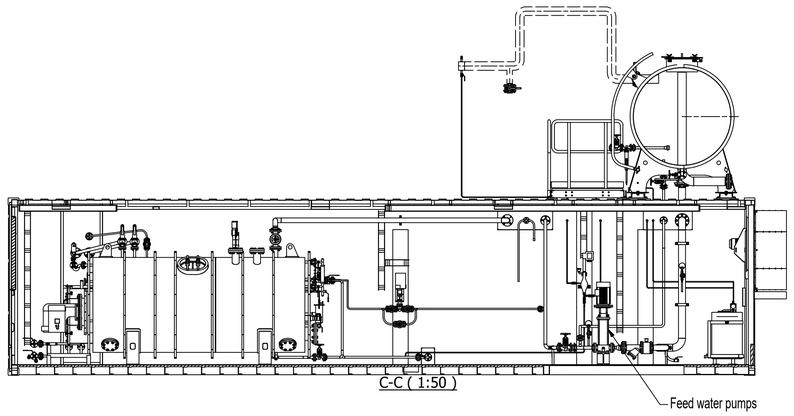


B (1:20)
All Flanges of Container Wall





A-A (1:50)



C-C (1:50)

LOADINGS

WIND LOADS ARE CALCULATED BY USING SAFETY MARGIN AND ON THE LEVEL +-0.000.
LOADS FROM TRAFFIC, SNOW ETC. ARE TO BE TAKEN INTO CONSIDERATION BY OTHERS.

WIND LOAD PER LEG OF CONTAINER
MAX. TOTAL SHEAR (kN): XX
MAX. UPLIFT FORCE (kN): XX
MAX. COMPRESSION FORCE (kN): XX
MAX. MOMENT OF STRUCTURE (kNm): XX

EQ LOAD PER LEG OF CONTAINER
MAX. TOTAL SHEAR (kN): XX
MAX. UPLIFT FORCE (kN): XX
MAX. COMPRESSION FORCE (kN): XX
MAX. MOMENT OF STRUCTURE (kNm): XX

WIND LOADS OF CHIMNEY - W1
MAX. VERTICAL FORCE (kN): XX
MAX. SHEAR FORCE (RESULTANT) (kN): XX
MAX. BENDING MOMENT (RESULTANT) (kNm): XX
W2 (kN): XX

EQ LOADS OF CHIMNEY - W1
MAX. VERTICAL FORCE (kN): XX
MAX. SHEAR FORCE (RESULTANT) (kN): XX
MAX. BENDING MOMENT (RESULTANT) (kNm): XX

SITE CONDITIONS ACC.TO UBC 1997		
Description	Value	Unit
WIND		
Basic wind speed	XX	mph
Exposure class	XX	
Wind importance factor	XX	
Pressure coefficients	XX	
EARTHQUAKE		
Seismic zone	XX	
Seismic source type	XX	
Min. distance to seismic source	XX	km
Soil profile type	XX	
Seismic importance factor	XX	
Response modification factor	XX	

DELIVERY LIMITS ARE SHOWN BY — ALFA LAVAL AALBORG OY / OTHERS

GENERAL TOLERANCE FOR TERMINAL POINTS: +-20

WEIGHT INFORMATION
DRY WEIGHT: 15500
MAX. WEIGHT: 25000
OPERATING WEIGHT: 18500

CONTAINER ASSEMBLY TO THE FOUNDATION BED:

CONTAINER SUPPORT LEGS (4 PCS) ARE FILLET WELDED AROUND TO THE FOUNDATION PLATES. ALTERNATIVE FIXING WITH CHEMICAL ANCHOR BOLTS.

IT IS RECOMMENDED THAT FOUNDATION BED IS MIN. 100 MM WIDER AND 100 MM LONGER THAN CONTAINER FRAME.
FOUNDATION PLATE IS TO BE MIN. 50 MM WIDER AND 50 MM LONGER THAN CONTAINER SUPPORT LEG.
FOUNDATION PLATES ARE TO BE BOLTED OR MOULDED TO THE FOUNDATION BED.
DURING INSTALLATION STAGE, THERMAL EXPANSIONS ARE TO BE TAKEN INTO THE CONSIDERATION;
SEE AALBORG INDUSTRIES GENERAL INSTRUCTIONS AND OTHER RELATED INSTALLATION DOCUMENTS.
DESIGN FOR THE FOUNDATION BED AND PLATES IS OUT OF ALFA LAVAL SCOPE OF SUPPLY.

THE SLIDING SUPPORTS AND FIXED POINTS OF THE PIPELINES
TO BE DESIGNED SO THAT THE MASS FORCES AND THERMAL
EXPANSIONS ARE COMPENSATED

Pos	Connection	Note	Size	PN
W	Exhaust Gas Outlet		DN250	
T	Alarms / Indications			
S	Power Supply	3x400V / 60Hz		
Q	Service/Cooling Water		DN15	40
G6	Water from Oil Detector		Ø12	
F1	Water to Oil Detector		Ø12	
G3	Chemical to Feed Water Tank		Ø10	
G2	Chemical to Feed Water Tank		Ø10	
G1	Pump min. flow to Feed Water Tank		DN25	40
G4	LFO Fuel Return		DN25	40
F2	LFO Fuel to Burner		DN25	40
D1	Venting		DN25	40
C3	Control Panel Drain		Ø12	
G5	Boiler Scum / Bottom BlowOut		DN25	40
C2	Floor Drain		2"	
C1	Floor Drain		2"	
B2	Steam to Consumers		DN65	16
B1	Feed Water to Boilers		DN40	40
Ae2	Safety Valve Blow Out		DN50	40
Ae1	Safety Valve Blow Out		DN50	40
A1	Feed Water to Feed Water Pump		DN100	16

Fax.	Flat.	Fvert.	Mtors.	Mlat.	Mvert.
Max. Loads (kN)		Max. Moments (kNm)			

ALLOWABLE FORCES AND MOMENTS
- Not to be subjected to any fluctuating loads.
- All loads are only static (+-).
- Forces given in the table are not allowed to be exceeded.
- Other pipelines to be designed so that the mass forces and thermal expansions are compensated.

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Power Plants

WÄRTSILÄ

Product WHR-Steam system

Main dimensions
Heat Recovery Container
RHC 901
8159 Mansourah - Massarah Gold Mine Proj no 8159

MADE	07/04/2020	INPANKRA	
CHKD			
APPD			

Units cm/kg | Weight 15293 kg | Customer Document
Page No 1 /1 | Scale 150 | Size A1 | Doc. ID 8159-94M | Rev

MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **Main Dimensions - CONDENSATE TANK UNIT, CU-1**
49PR POWER GENERATION
Dimensional drawing

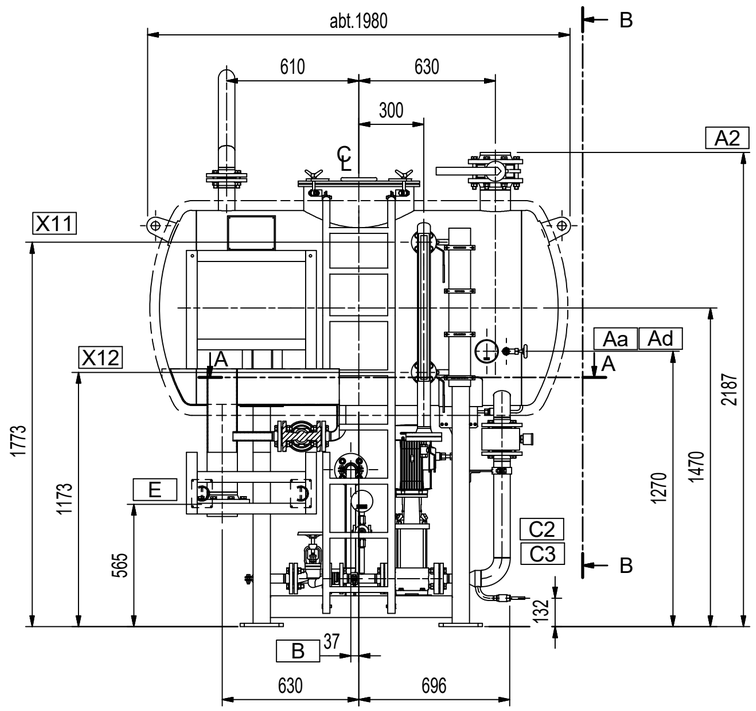
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Revision Date:
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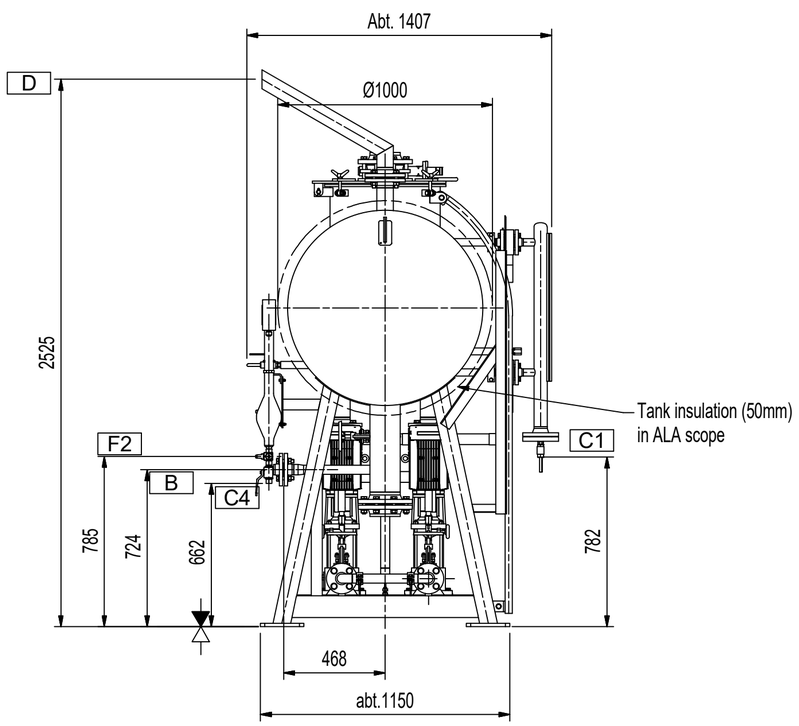
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L&T Document Number:							

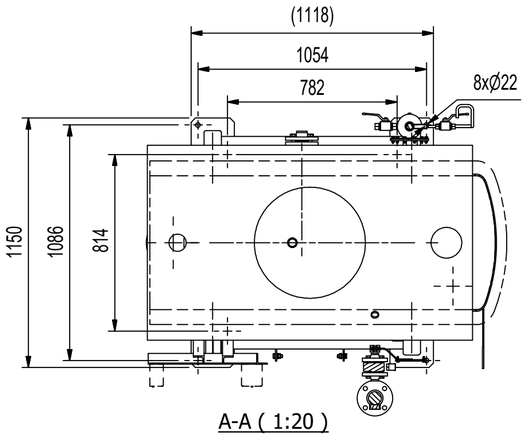
WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF490873 a
	Original Doc no: 8159-101M
	Wartsila Supplier:

REVISION HISTORY

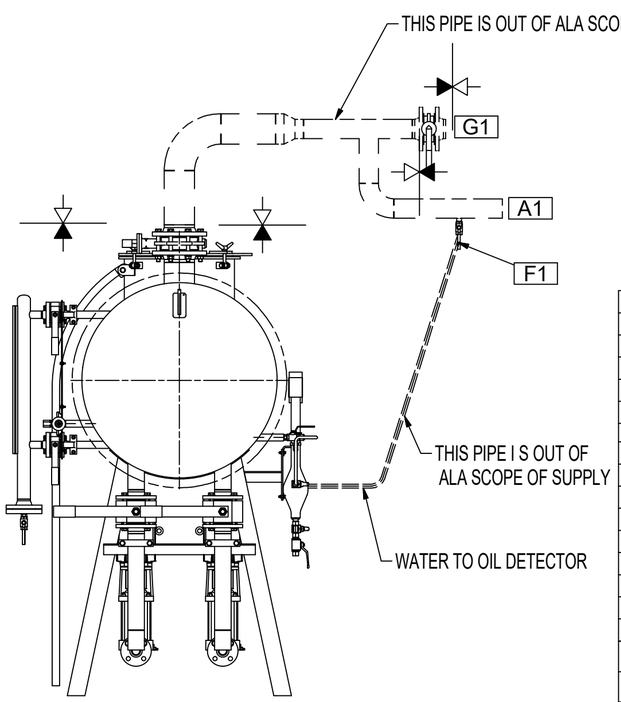
Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer



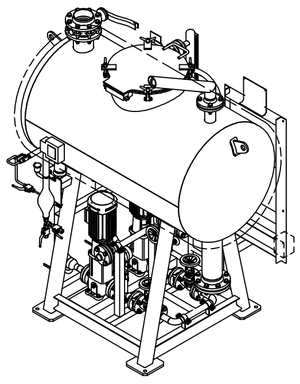


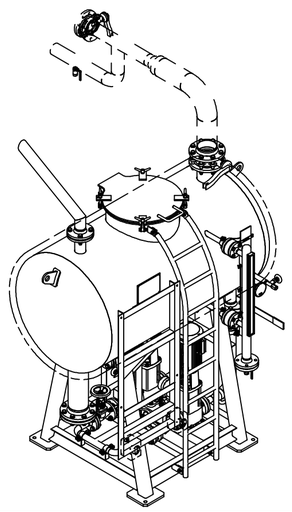


A-A (1:20)



B (1:20)





ASSEMBLY TOLERANCES FOR FOUNDATIONS	
THE FOLLOWING GENERAL TOLERANCES ARE TO BE FOLLOWED DURING ASSEMBLY OF BASE PLATES OR ANCHOR BOLTS TO THE CONCRETE BED:	
CENTER LINE DISTANCE INSIDE ONE ANCHOR BOLT GROUP	±2
DIAGONAL DIMENSION INSIDE ONE ANCHOR BOLT GROUP	±3
ELEVATION POSITION FOR ANCHOR BOLTS	±10
CENTER LINE DISTANCE BETWEEN ANCHOR BOLT GROUPS	±5
STRAIGHTNESS OF BASE PLATES	±2
CENTER LINE DISTANCE BETWEEN BASE PLATES	±5
ELEVATION POSITION FOR BASE PLATES	±5

GENERAL TOLERANCE FOR TERMINAL POINTS: ±20

THE SLIDING SUPPORTS AND FIXED POINTS OF THE PIPELINES TO BE DESIGNED SO THAT THE MASS FORCES AND THERMAL EXPANSIONS ARE COMPENSATED.

THE ELEVATION OF THE FEED WATER TANK IS TO BE DETERMINED ACCORDING TO F.W. PUMP'S NPSH REQUIRED VALUE.
SEE FURTHER INFORMATION FROM PUMP'S DATA SHEET.

WEIGHT INFORMATION

DRY WEIGHT:	860 kg
MAX. WEIGHT:	1850 kg
OPERATING WEIGHT:	1350 kg

FOUNDATION LOADING

Fst = kN STATIC LOAD IN DIRECTION Z

NOTE!
LOADS ARE CALCULATED BY USING SAFETY COEFFICIENT VALUE 1.
LOADS FROM TRAFFIC, SNOW ETC. ARE TO BE TAKEN INTO THE CONSIDERATION BY OTHERS.

DELIVERY LIMITS ARE SHOWN BY — ALFA LAVAL AALBORG OY / OTHERS

UNIT ASSEMBLY TO THE FOUNDATION BED:
1. WITH ANCHOR BOLTS. CHEMICAL ANCHOR BOLTS ARE RECOMMENDED.
2. UNIT LEGS ARE FILLET WELDED AROUND TO THE FOUNDATION PLATES.

FOUNDATION BED IS TO BE MIN. 100 MM WIDER AND 100 MM LONGER THAN UNIT FRAME.
FOUNDATION PLATE IS TO BE MIN. 50 MM WIDER AND 50 MM LONGER THAN UNIT LEG.
FOUNDATION PLATES ARE TO BE BOLTED OR MOULDED TO THE FOUNDATION BED.
DURING INSTALLATION STAGE, THERMAL EXPANSIONS ARE TO BE TAKEN INTO THE CONSIDERATION;
SEE ALFA LAVAL AALBORG GENERAL INSTRUCTIONS.
DELIVERY AND DESIGN FOR THE FOUNDATION BED AND PLATES IS OUT OF ALFA LAVAL AALBORG SCOPE OF SUPPLY.

Fx	Fy	Fz	Mx	My	Mz	Pos	Connection	Note	Size	PN
Max. Loads (kN)			Max. Moments (kNm)							
x	x	x	x	x	x	X12	LOCAL LEVEL INDICATIONS		DN25	PN16
x	x	x	x	x	x	X11	LOCAL LEVEL INDICATIONS		DN25	PN16
x	x	x	x	x	x	G1	OILY CONDENSATE		DN80	PN16
x	x	x	x	x	x	F2	WATER TO OIL DETECTOR		G½"	PN16
x	x	x	x	x	x	F1	WATER TO OIL DETECTOR		G½"	PN16
0,2	0,2	0,2	0,08	0,08	0,08	E	OVERFLOW / DRAIN	CLEAN	DN125	PN16
						D	VENTING		DN65	PN16
x	x	x	x	x	x	C4	DRAIN	DIRTY	G½"	PN16
x	x	x	x	x	x	C3	DRAIN	CLEAN	G½"	PN16
x	x	x	x	x	x	C2	DRAIN	CLEAN	G½"	PN16
x	x	x	x	x	x	C1	WATER FROM WATER LEVEL GAUGE	CLEAN	G 1/2"	PN16
0,1	0,1	0,1	0,02	0,02	0,02	B	CONDENSATE TO FEED WATER TANK		DN40	PN16
0,3	0,3	0,3	0,1	0,1	0,1	A2	CONDENSATE IN		DN125	PN16
x	x	x	x	x	x	A1	CONDENSATE FROM OWN CONSUMPTION		DN80	PN16
x	x	x	x	x	x	Ad	SAMPLE		G 1/2"	PN16
x	x	x	x	x	x	Aa	LOCAL TEMPERETURE INDICATION		G 1/2"	PN16

ALLOWABLE FORCES AND MOMENTS

- Not to be subjected to any fluctuating loads.
- All loads are only static (+-).
- Forces given in the table are not allowed to be exceeded.
- Other pipelines to be designed so that the mass forces and thermal expansions are compensated.

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Power Plants

WÄRTSILÄ

ALFA LAVAL

Product WHR-Steam system | ACA | ASLY.DRG

Main dimensions

CONDENSATE TANK UNIT, CU-1

RDD 901

8159 Mansourah - Massarah Gold Mine Proj. no. 8159

MADE	7.4.2020	FIRAJSNB	Units: mm/kg
CHKD	Ref. to:	FIRA:QMS (2018)	Page No. 1 /1
APPD	Ref. to:	FIRA:QMS (2018)	

Weight 848 kg | Customer Document: -

Scale: 1:20 | Size: A2 | Doc. ID: 8159-101M | Rev. -

Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **Main dimensions - BLOW DOWN TANK, BDT-150**
49PR POWER GENERATION
Dimensional drawing

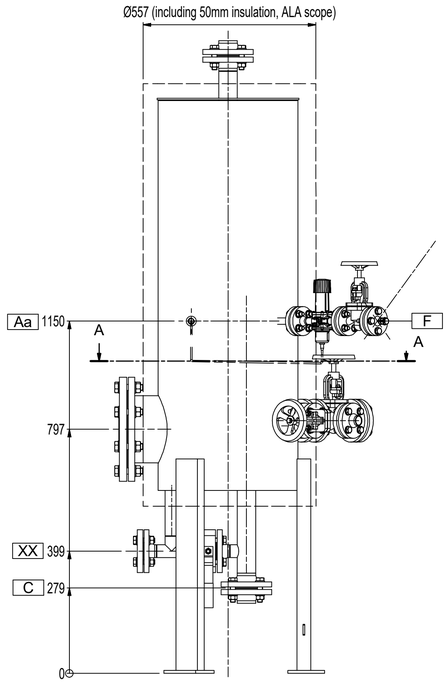
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Revision:
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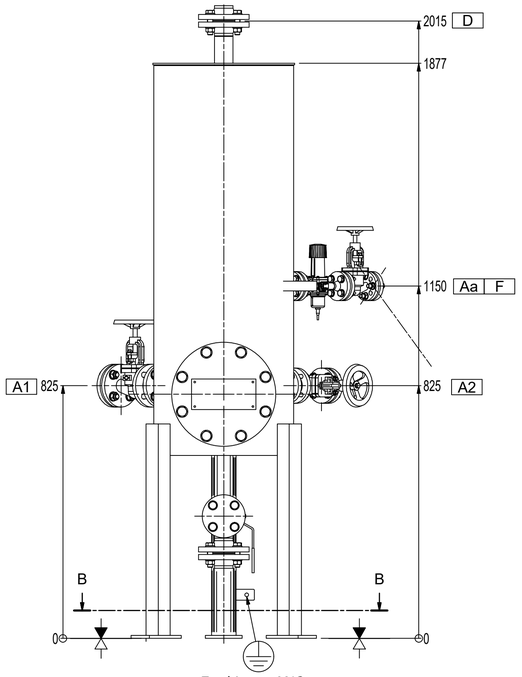
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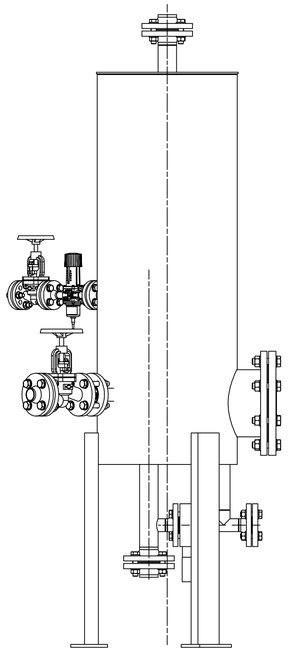
WÄRTSILÄ	Wartsila Project ID: P/20002
	Wartsila Doc no: DBAF490875 a
	Original Doc no: 8159-113M
	Wartsila Supplier:

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer







ASSEMBLY TOLERANCES FOR FOUNDATIONS	
THE FOLLOWING GENERAL TOLERANCES ARE TO BE FOLLOWED DURING ASSEMBLY OF BASE PLATES OR ANCHOR BOLTS TO THE CONCRETE BED:	
CENTER LINE DISTANCE INSIDE ONE ANCHOR BOLT GROUP	±2
DIAGONAL DIMENSION INSIDE ONE ANCHOR BOLT GROUP	±3
ELEVATION POSITION FOR ANCHOR BOLTS	±10
CENTER LINE DISTANCE BETWEEN ANCHOR BOLT GROUPS	±5
STRAIGHTNESS OF BASE PLATES	±2
CENTER LINE DISTANCE BETWEEN BASE PLATES	±5
ELEVATION POSITION FOR BASE PLATES	±5

GENERAL TOLERANCE FOR TERMINAL POINTS: ±20

WEIGHT INFORMATION

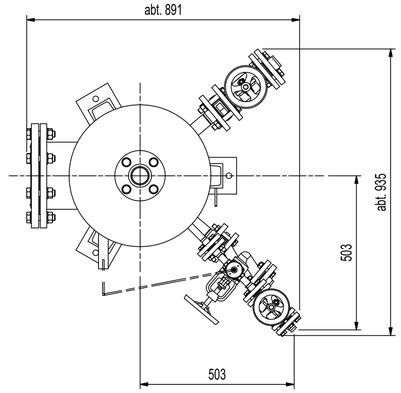
DRY WEIGHT:	210 kg
MAX. WEIGHT:	390 kg
OPERATING WEIGHT:	320 kg

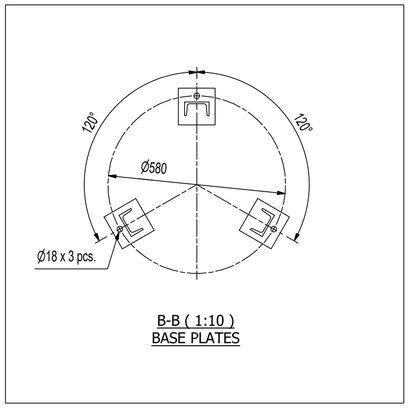
THE SLIDING SUPPORTS AND FIXED POINTS OF THE PIPELINES TO BE DESIGNED SO THAT THE MASS FORCES AND THERMAL EXPANSIONS ARE COMPENSATED.

UNIT ASSEMBLY TO THE FOUNDATION BED:
1. WITH ANCHOR BOLTS. CHEMICAL ANCHOR BOLTS ARE RECOMMENDED.
2. UNIT LEGS ARE FILLET WELDED AROUND TO THE FOUNDATION PLATES.

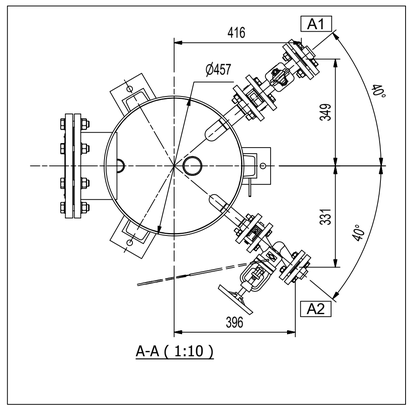
FOUNDATION BED IS TO BE MIN. 100 MM WIDER AND 100 MM LONGER THAN UNIT FRAME.
FOUNDATION PLATE IS TO BE MIN. 50 MM WIDER AND 50 MM LONGER THAN UNIT LEG.
FOUNDATION PLATES ARE TO BE BOLTED OR MOULDED TO THE FOUNDATION BED.
DURING INSTALLATION STAGE, THERMAL EXPANSIONS ARE TO BE TAKEN INTO THE CONSIDERATION; SEE ALFA LAVAL AALBORG GENERAL INSTRUCTIONS.
DELIVERY AND DESIGN FOR THE FOUNDATION BED AND PLATES IS OUT OF ALFA LAVAL AALBORG SCOPE OF SUPPLY.

DELIVERY LIMITS ARE SHOWN BY — ALFA LAVAL AALBORG OY / OTHERS

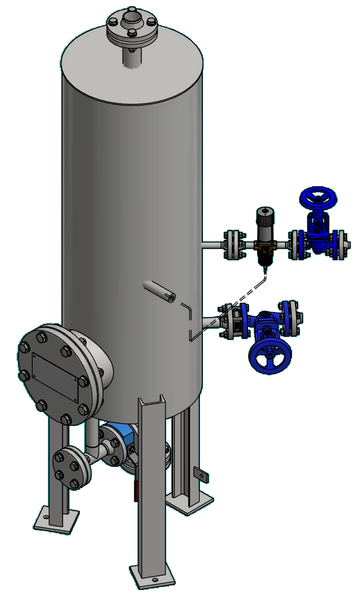




B-B (1:10)
BASE PLATES



A-A (1:10)



x	x	x	x	x	x	XX	CLEANING CONNECTION		DN32	PN40
0,1	0,1	0,1	0,02	0,02	0,02	F	COOLING WATER IN		DN15	PN40
0,1	0,1	0,1	0,02	0,02	0,02	D	VENTING		DN50	PN40
0,1	0,1	0,1	0,02	0,02	0,02	C	OVERFLOW / DRAIN		DN50	PN40
x	x	x	x	x	x	Aa	COOLING SENSOR		G1/2" female	
0,2	0,2	0,2	0,035	0,035	0,035	A2	BLOW OUT FROM AUX. BOILER		DN25	PN40
0,2	0,2	0,2	0,05	0,05	0,05	A1	BLOW OUT FROM BOILERS		DN32	PN40
Fx	Fy	Fz	Mx	My	Mz	Pos	Connection	Note	Size	PN
Max. Loads (kN)			Max. Moments (kNm)							

ALLOWABLE FORCES AND MOMENTS
- Not to be subjected to any fluctuating loads.
- All loads are only static (+-).
- Forces given in the table are not allowed to be exceeded.
- Other pipelines to be designed so that the mass forces and thermal expansions are compensated.

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Power Plants

WÄRTSILÄ

Main dimensions

BLOW DOWN TANK, BDT-150

Product WHR-Steam system | ACA | ASLY.DRG

8159 Mansourah - Massarah Gold Mine Proj. no. 8159

MADE	2.4.2020	FIRAJSNB	Units: mm/kg	Weight 244 kg	Customer Document: -
CHKD	Ref. to:	FIRA:QMS (2018)	Page No. 1 /1	Scale: 1:10	Size: A1
APPD	Ref. to:	FIRA:QMS (2018)		Doc. ID: 8159-113M	Rev. -

MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **Dimensional drawing**
49PR POWER GENERATION
Boiler Washing Water Tank

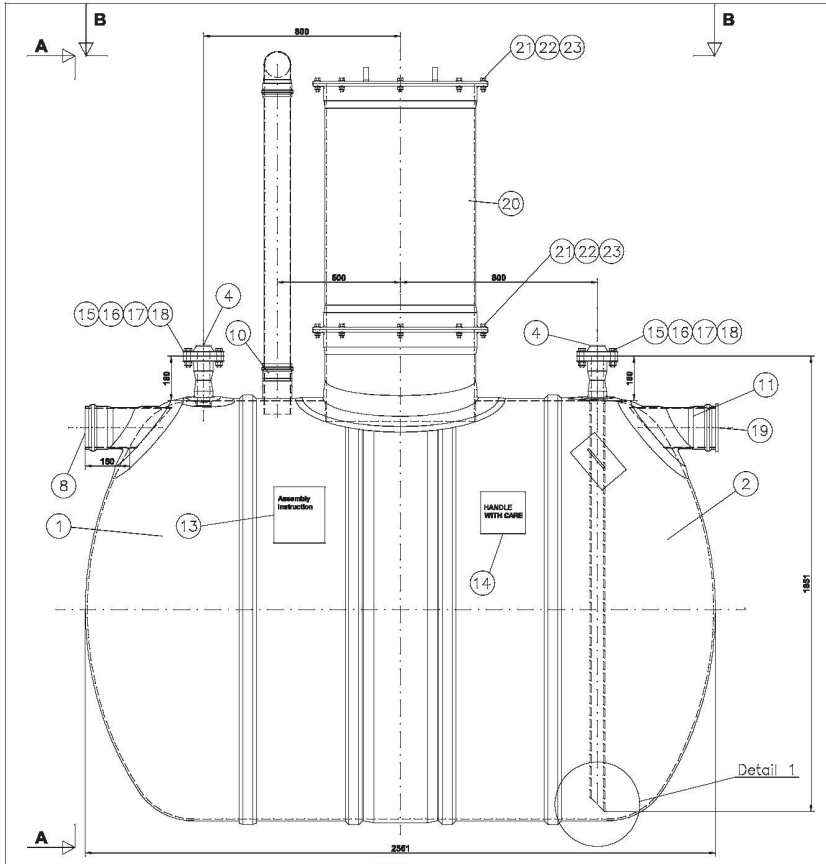
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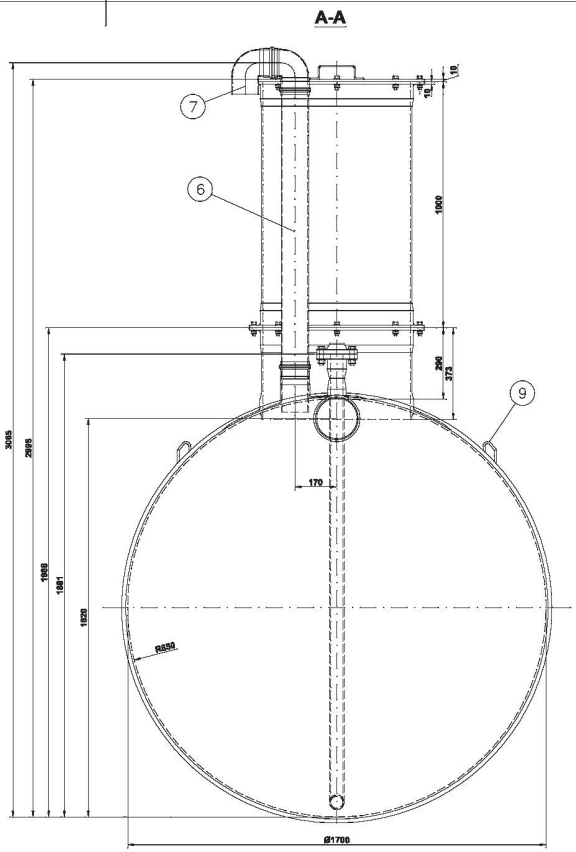
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L&T Document Number:							

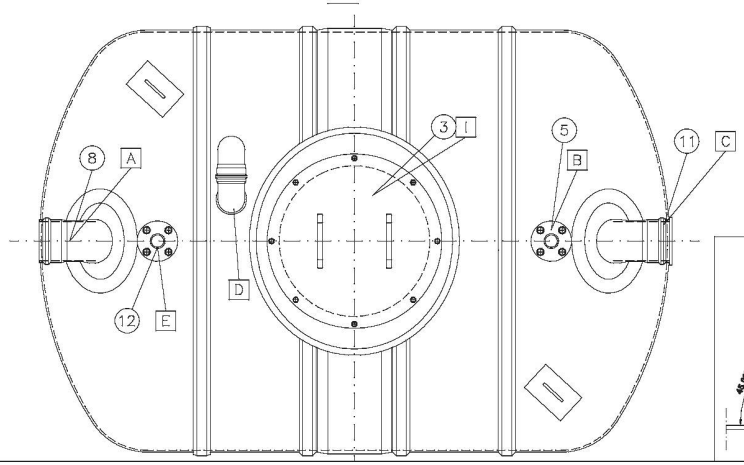
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	Wartsila Doc no: DBAF508721 -
	Original Doc no: 1-16110200 A
	Wartsila Supplier:

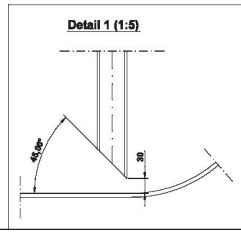
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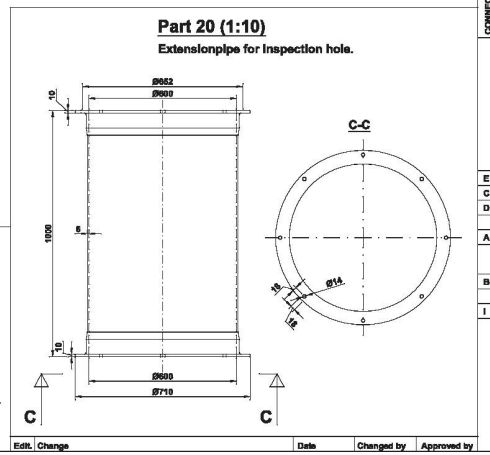
Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer







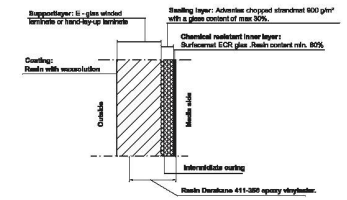




TECHNICAL DATA	
CONTENTS	
DENSITY OF CONTENTS	
OPERATING TEMPERATURE	80°C
DESIGN TEMPERATURE	85°C
DISIGN PRESSURE	0.2 bar
TOTAL VOLYM	5.12 m ^{3}
MATERIAL	GRP
WEIGHT	500 kg
STUFFED	
INSULATION	

TECHNICAL DATA

DESIGN



SYMBOLS

SM	= Surfacemat Owens Corning M524 - ECR30S, 30g/m²
M300	= Chopped strand mat Owens Corning M723A, 300g/m²
M450	= Chopped strand mat Owens corning M723A, 450g/m²
R600	= 1/1 Roving Dipex STO 600-1250, 600g/m²
U460	= Winding roving, Owens Corning R25HX14 2400 C-F 24MIC
UD500	= Axial roving, Ahlstrom 9887B-25-225 R24-500-T, 500g/m²
Combi	= Owens Corning LT 0600 / S 300/06 H 01/125, 900 g/m²
411-350	= Derakane 411-350 epoxy vinylester resin. HDT 105°

Assembly depth maximum 1 meter below ground level.

For the installation 2 pcs off anchoring belts are included in the delivery off the tank.

All connections are plugged during transport.

CONNECTION CODE	Item/ref	Quantity	Title/Name, designation, material, dimension etc	Material	Standard	PCS	Notice
	23	Washer M12		FeZnk	DIN 125	32	
	22	Nut M12		FeZnk	DIN 934	16	
	21	Bolt M12x45		FeZnk	DIN 933	16	
	20	Extensionpipe for the inspectionhole		GRP		1	See detail
	19	Plugg Ø 160		PVC		1	
	18	Washer M 16		FeZnk	DIN 125	16	
	17	Nut M 16		FeZnk	DIN 934	8	
	16	Bolt M16x70		FeZnk	DIN 931	8	
	15	Gasket DN 50		EPDM		2	
	14	Handel with care sign.				1	
	13	Sign with assembly instruction				1	
E	12	Collar DN50 PN10 (Extra)		GRP		1	counter flange
C	11	PVC sleeve joint DI 160mm (inlet)		PVC		1	L=320, plugged
D	10	PVC sleeve joint DI 110mm		PVC		1	L=200
	9	Lifting hook		S235JRG2		2	
A	8	PVC sleeve joint DI 160mm (inlet)		PVC		1	L=320
	7	Bend for venting pipe DI 110mm		PVC		2	
	6	Venting pipe DI 110mm		PVC		1	L=1200
B	5	Collar DN50 PN10 (Outlet=suctionpipe)		GRP		1	counter flange
	4	Weld-Neck Flange DN 50 PN16		S235JRG2	DIN 2633 304 L	2	
I	3	Manhole DN600 PN6		GRP		1	
	2	Gable Ø1700, 9.0 mm		GRP		1	
	1	Gable Ø1700, 9.0 mm		GRP		1	

Edit.	Change	Date	Changed by	Approved by

Designed by	Checked by	Approved by – date	Filename	Date	Scale
JKI			151016	15.11.2016	1/10

NCE Nordjysk Consulting Engineering A/S HEJMDAL

Boiler washing water tank under ground

DRG.No. 1-16110200 | Edition A | Sheet 1/1

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	01-12-2020	07-01-2021	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	01-12-2020	07-01-2021	26-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
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ATTACHMENT 3 REFERENCE FOR SUMP PIT DESIGN

Rectangular Tanks

32.1 Introduction. Rectangular tanks are normally provided when capacity of the liquid to be stored is small. For small capacities, circular tanks work out to be uneconomical on account of high cost of curved shuttering. On the other hand, circular tanks by virtue of their having least perimeter for a given capacity, require least amount of materials and as such prove to be economical for large capacities of storage. However in case of circular tanks, it is difficult to divide the tank in compartments and they do not occupy the entire available area of land. Rectangular tanks do not have such disadvantage.

The components of a rectangular tank are :

1. Side walls
2. Base slab
3. Roof slab

32.2 Design of side walls. The walls of a rectangular tank are liable to bend both in the horizontal as well as vertical direction. The method to be adopted for designing the wall panels depend upon the relative proportions of the length, breadth and height of the tank. The conditions of support provided to the tank wall at top and bottom also influences the method of design. Since the water pressure acts in the form of triangular load on the walls, the exact analysis becomes rather difficult and as such the walls are normally designed by approximate method of design.

Let L = length of the tank

B = width of the tank

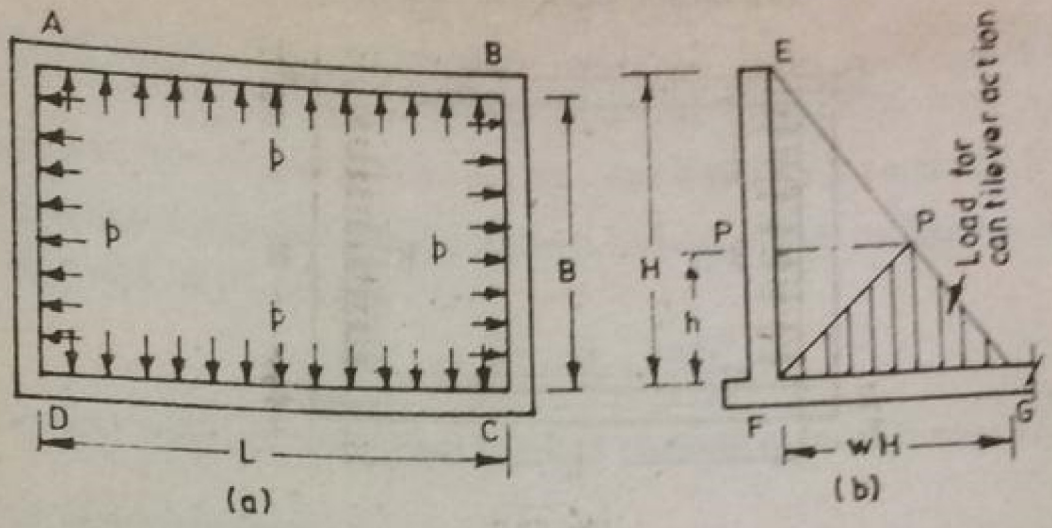
and H = height of the tank

The design of the walls by approximate method can be broadly divided in two categories.

(*a*) Tank wall having ratio of $\frac{L}{B}$ less than 2

(*b*) Tank wall having ratio of $\frac{L}{B}$ greater than 2

(*a*) **Tank wall having ratio of $\frac{L}{B} < 2$.** For rectangular tank in whic ratio of length to width is less than 2, the tank walls are designed as horizont slabs, continuous all round (between corners) and subjected to triangular lo



Plan of tank showing pressure acting on walls

Load Distribution on wall

Fig. 32.1

due to water pressure varying from zero at top to a maximum at a height of h = $\frac{H}{4}$ or 1 m above base whichever is more. The bottom portion for a height h = $\frac{H}{4}$ or 1 m (whichever is more), the wall is treated as vertical cantilever fixed base, and subjected to triangular loading given by the area P.F.G. of t pressure diagram (*i.e.* varying from zero at P to wH at base).

Ref. Fig. 32.1 maximum pressure (p) for design of wall for horizon bending per unit height at level of P

$$= w\,(H - h) \text{ per m}$$

The moments can be easily found by moment distribution.

In case moment distribution is not carried out, the following values B.M. can be adopted in the design.

B.M. at end of span $= \frac{pB^2}{12}$ (producing tension on water force)

B.M. at centre of span

$$= \frac{pB^2}{16} \text{ (producing tension on outer face).}$$

Pull in long and short walls. Due to monolithic construction, in add to bending moments, both the long as well as the short walls are also subj to direct tension or pull caused by water pressure on wall.

(*a*) *Direct tension or pull on long walls.* Since the short walls between the long wall, above P, the water pressure on short wall transformed to the long wall as tension. Consider a 1 m high strip of shor at a level h $\left(= 1\text{m or } \frac{H}{4} \text{whichever is more} \right)$ above base *i.e.* at P.

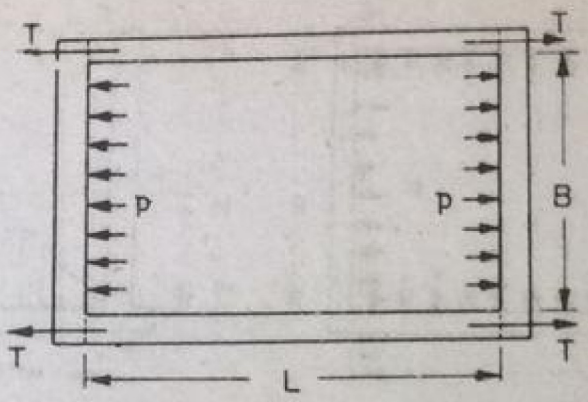


Fig. 32.2
Plan showing pressure causing pull in long walls

Total tension due to water pressure

$$= w\,(H - h) \times 1 \times B$$

$\therefore$ Direct tension or pull transferred to each long wall

$$= T_L = \frac{1}{2} \cdot w \cdot (H - h)\, B$$

Similarly

(b) Direct tension or pull on short walls

$$= T_B = \frac{1}{2} \cdot w \cdot (H - h) \cdot L$$

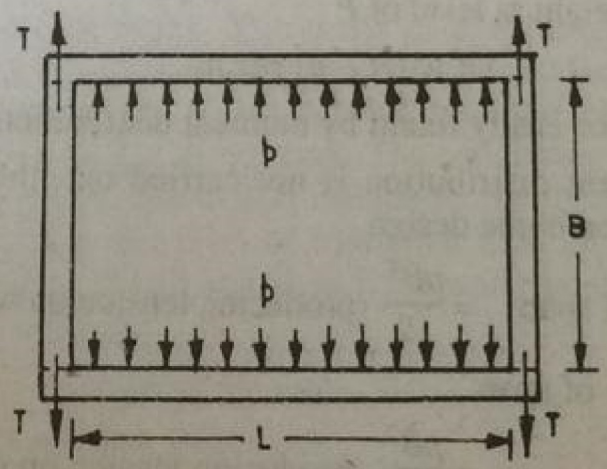


Fig. 32.3
Plan showing pressure causing pull in short walls

Cantilever moment in short walls :

Maximum cantilever B.M. in short walls

$$= wH \cdot \frac{h}{2} \cdot \frac{h}{3} = \frac{wH \cdot h^2}{6}$$

(*b*) Tank wall having ratio of $\frac{L}{B} > 2$. For rectangular tanks in which ratio of length to width is more than 2, the long walls are treated as